

1. action

The action namespace is produced by one regex pass over one combined text string: tweet body + OCR text + transcript text + media-description text, joined with separators. A tag is added when its regex appears anywhere in that combined string.

1.1. action:detention (2,278 (22.8%))

Rule: Matches arrest/arrested/arresting, detain/detained/detaining, apprehend/apprehended/apprehending, in custody, or nab/nabbed.

```
518: PATTERN_ACTION_DETENTION = _compile(  
519:     r"\b(arrest(?:ed|ing)?|detain(?:ed|ing)?|apprehend(?:ed|ing)?|in custody|nab(?:bed)?)\b"  
520: )  
  
1540:     (PATTERN_ACTION_DETENTION, "action:detention"),
```

1.2. action:deportation (1,304 (13.0%))

Rule: Matches deport/deported/deporting/deportation, remove/removed/removal, or repatriated/repatriation, but the deport match is guarded so self-deport does not count here.

```
528: PATTERN_ACTION_DEPORTATION = _compile(  
529:     r"(?<self-)(?!self)\b(deport(?:ed|ing|ation)?|remov(?:ed|al)|repatriat(?:ed|ion))\b"  
530: )  
  
1542:     (PATTERN_ACTION_DEPORTATION, "action:deportation"),
```

1.3. action:self-deportation (191 (1.9%))

Rule: Matches self-deport/self deport variants, voluntary/voluntarily departure/departed/departing, leave voluntarily, report <your/their/his/her/an/the> departure, or take control of <your/their/his/her> departure.

```
521: PATTERN_ACTION_SELF_DEPORTATION = _compile(  
522:     r"\bself[- ]deport(?:ed|ing|ation)?\b"  
523:     r"|\bvolutar(?:y|ily)\s+depart(?:ure|ed|ing)?\b"  
524:     r"|\bleave voluntarily\b"  
525:     r"|\breport\s+(?:your|their|his|her|an|the)?\s*departure\b"  
526:     r"|\btake\s+control\s+of\s+(?:your|their|his|her)\s+departure\b"  
527: )  
  
1541:     (PATTERN_ACTION_SELF_DEPORTATION, "action:self-deportation"),
```

1.4. action:report-immigrants (7 (0.1%))

Rule: Matches report/call/tip language only when ICE/DHS/contact information and immigrant/alien/migrant/immigration-violation subject language occur in the same bounded expression. The bounds in the regex are 80, 120, 180, or 240 characters depending on phrase order.

```
546: PATTERN_ACTION_REPORT_TO_ICE = _compile(  
547:     rf"\b(?:"  
548:     rf"(?:call|contact)\s+(?:{ICE_DIRECT_TARGET}|{ICE_REPORT_PHONE})\b"  
549:     rf"(?:={0,180})\b{ICE_REPORT_SUBJECT}\b)"  
550:     rf"|(?:report|tip|submit|send|notify)\b.{0,80}\b{ICE_REPORT_SUBJECT}\b"  
551:     rf"|.{{0,80}}\b(?:to|at|with)\s+(?:{ICE_OR_DHS_TARGET}|{ICE_REPORT_PHONE})\b"  
552:     rf"|(?:submit|send)\s+(?:a\s+)?tip\b.{0,80}\b(?:to|with)\s+"  
553:     rf"(?:{ICE_OR_DHS_TARGET}|{ICE_REPORT_PHONE})\b"  
554:     rf"(?:={0,120})\b{ICE_REPORT_SUBJECT}\b)"  
555:     rf"|\b{ICE_REPORT_SUBJECT}\b.{0,240}\b(?:call|contact)\s+"  
556:     rf"(?:our|the)\s+(?:{ICE_DIRECT_TARGET}|ICE\s+)"  
557:     rf"(?:tip\s*line|hotline|{ICE_REPORT_PHONE})\b"  
558:     rf>")"  
559: )  
  
1543:     (PATTERN_ACTION_REPORT_TO_ICE, "action:report-immigrants"),
```

2. artist

The artist namespace is produced only from written or transcribed text. The code does not identify music by listening to audio. It matches artist names or listed signature song phrases in the combined text string.

2.1. artist:lee-greenwood (4 (0.0%))

Rule: Matches Lee Greenwood, God Bless the USA, or Proud to be an American in written/transcribed text.

```

941: ARTIST_GAZETTEER: tuple[tuple[re.Pattern[str], str], ...] = (
942:     (_compile(r"\bsydney\s+sweeney\b"), "sydney-sweeney"),
943:     (
944:         _compile(
945:             r"\blee\s+greenwood\b|\bgod\s+bless\s+the\s+u\.?s\.?a\.?b"
946:             r"|\bproud\s+to\s+be\s+an\s+american\b"
947:         ),
948:         "lee-greenwood",
949:     ),
950:     (_compile(r"\bkid\s+rock\b"), "kid-rock"),
951:     (_compile(r"\bvillage\s+people\b|\bmacho\s+man\b"), "village-people"),
952:     (_compile(r"\bjason\s+aldean\b|\btry\s+that\s+in\s+a\s+small\s+town\b"), "jason-aldean"),
953:     (_compile(r"\boliver\s+anthony\b|\brich\s+men\s+north\s+of\s+richmond\b"), "oliver-anthony"),
954:     (_compile(r"\btaylor\s+swift\b"), "taylor-swift"),
955:     (_compile(r"\bbeyonc[eé]\b"), "beyonce"),
956: )

1672: # artist:<name> – named cultural figures / musicians (and signature songs).
1673: for artist_pat, artist_slug in ARTIST_GAZETTEER:
1674:     if m := artist_pat.search(text):
1675:         add(f"artist:{artist_slug}", span=m.span())

```

2.2. artist:kid-rock (2 (0.0%))

Rule: Matches Kid Rock in written/transcribed text.

```

941: ARTIST_GAZETTEER: tuple[tuple[re.Pattern[str], str], ...] = (
942:     (_compile(r"\bsydney\s+sweeney\b"), "sydney-sweeney"),
943:     (
944:         _compile(
945:             r"\blee\s+greenwood\b|\bgod\s+bless\s+the\s+u\.?s\.?a\.?b"
946:             r"|\bproud\s+to\s+be\s+an\s+american\b"
947:         ),
948:         "lee-greenwood",
949:     ),
950:     (_compile(r"\bkid\s+rock\b"), "kid-rock"),
951:     (_compile(r"\bvillage\s+people\b|\bmacho\s+man\b"), "village-people"),
952:     (_compile(r"\bjason\s+aldean\b|\btry\s+that\s+in\s+a\s+small\s+town\b"), "jason-aldean"),
953:     (_compile(r"\boliver\s+anthony\b|\brich\s+men\s+north\s+of\s+richmond\b"), "oliver-anthony"),
954:     (_compile(r"\btaylor\s+swift\b"), "taylor-swift"),
955:     (_compile(r"\bbeyonc[eé]\b"), "beyonce"),
956: )

1672: # artist:<name> – named cultural figures / musicians (and signature songs).
1673: for artist_pat, artist_slug in ARTIST_GAZETTEER:
1674:     if m := artist_pat.search(text):
1675:         add(f"artist:{artist_slug}", span=m.span())

```

3. country

The country namespace is produced by validated extraction. The code first captures a one- or two-word place after one of these prepositions: from, in, to, of, with, by. It then resolves that capture against the country vocabulary. It also adds country tags for curated demonyms and curated foreign city names.

3.1. country:Mexico (771 (7.7%))

Rule: Emitted as country:Mexico when a validated candidate resolves to Mexico: a country name after from/in/to/of/with/by, a curated demonym for Mexico, or a curated foreign city mapped to Mexico.

```

153:      "Mexico",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.2. country:El-Salvador (266 (2.7%))

Rule: Emitted as country:El-Salvador when a validated candidate resolves to El Salvador: a country name after from/in/to/of/with/by, a curated demonym for El Salvador, or a curated foreign city mapped to El Salvador.

```

98:      "El Salvador",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.3. country:Guatemala (189 (1.9%))

Rule: Emitted as country:Guatemala when a validated candidate resolves to Guatemala: a country name after from/in/to/of/with/by, a curated demonym for Guatemala, or a curated foreign city mapped to Guatemala.

```

113:      "Guatemala",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.4. country:Honduras (180 (1.8%))

Rule: Emitted as country:Honduras when a validated candidate resolves to Honduras: a country name after from/in/to/of/with/by, a curated demonym for Honduras, or a curated foreign city mapped to Honduras.

```

117:      "Honduras",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>," - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.5. country:iran (133 (1.3%))

Rule: Emitted as country:iran when a validated candidate resolves to Iran: a country name after from/in/to/of/with/by, a curated demonym for Iran, or a curated foreign city mapped to Iran.

```

122:      "Iran",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.6. country:Venezuela (133 (1.3%))

Rule: Emitted as country:Venezuela when a validated candidate resolves to Venezuela: a country name after from/in/to/of/with/by, a curated demonym for Venezuela, or a curated foreign city mapped to Venezuela.

```

225:     "Venezuela",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*(?)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.7. country:Cuba (100 (1.0%))

Rule: Emitted as country:Cuba when a validated candidate resolves to Cuba: a country name after from/in/to/of/with/by, a curated demonym for Cuba, or a curated foreign city mapped to Cuba.


```

88:      "Cuba",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `*_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}}\s+({_PLACE}))\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.8. country:China (98 (1.0%))

Rule: Emitted as country:China when a validated candidate resolves to China: a country name after from/in/to/of/with/by, a curated demonym for China, or a curated foreign city mapped to China.

```

82:      "China",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.9. country:Dominican-Republic (70 (0.7%))

Rule: Emitted as country:Dominican-Republic when a validated candidate resolves to Dominican Republic: a country name after from/in/to/of/with/by, a curated demonym for Dominican Republic, or a curated foreign city mapped to Dominican Republic.

```

95:     "Dominican Republic",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     ""Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.10. country:Colombia (54 (0.5%))

Rule: Emitted as country:Colombia when a validated candidate resolves to Colombia: a country name after from/in/to/of/with/by, a curated demonym for Colombia, or a curated foreign city mapped to Colombia.

```

83:      "Colombia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.11. country:Laos (51 (0.5%))

Rule: Emitted as country:Laos when a validated candidate resolves to Laos: a country name after from/in/to/of/with/by, a curated demonym for Laos, or a curated foreign city mapped to Laos.

```

136:     "Laos",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.12. country:Ecuador (49 (0.5%))

Rule: Emitted as country:Ecuador when a validated candidate resolves to Ecuador: a country name after from/in/to/of/with/by, a curated demonym for Ecuador, or a curated foreign city mapped to Ecuador.

```

96:      "Ecuador",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.13. country:Israel (43 (0.4%))

Rule: Emitted as country:Israel when a validated candidate resolves to Israel: a country name after from/in/to/of/with/by, a curated demonym for Israel, or a curated foreign city mapped to Israel.

```

125:     "Israel",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.14. country:Vietnam (39 (0.4%))

Rule: Emitted as country:Vietnam when a validated candidate resolves to Vietnam: a country name after from/in/to/of/with/by, a curated demonym for Vietnam, or a curated foreign city mapped to Vietnam.

```

226:     "Vietnam",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.15. country:India (36 (0.4%))

Rule: Emitted as country:India when a validated candidate resolves to India: a country name after from/in/to/of/with/by, a curated demonym for India, or a curated foreign city mapped to India.


```

120:     "India",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.16. country:Somalia (35 (0.3%))

Rule: Emitted as country:Somalia when a validated candidate resolves to Somalia: a country name after from/in/to/of/with/by, a curated demonym for Somalia, or a curated foreign city mapped to Somalia.

```

192:     "Somalia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.17. country:Haiti (33 (0.3%))

Rule: Emitted as country:Haiti when a validated candidate resolves to Haiti: a country name after from/in/to/of/with/by, a curated demonym for Haiti, or a curated foreign city mapped to Haiti.

```

116:      "Haiti",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.18. country:Brazil (31 (0.3%))

Rule: Emitted as country:Brazil when a validated candidate resolves to Brazil: a country name after from/in/to/of/with/by, a curated demonym for Brazil, or a curated foreign city mapped to Brazil.

```

72:      "Brazil",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.19. country:Canada (31 (0.3%))

Rule: Emitted as country:Canada when a validated candidate resolves to Canada: a country name after from/in/to/of/with/by, a curated demonym for Canada, or a curated foreign city mapped to Canada.

```

79:      "Canada",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.20. country:Afghanistan (27 (0.3%))

Rule: Emitted as country:Afghanistan when a validated candidate resolves to Afghanistan: a country name after from/in/to/of/with/by, a curated demonym for Afghanistan, or a curated foreign city mapped to Afghanistan.

```

50:      "Afghanistan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.21. country:Jamaica (25 (0.2%))

Rule: Emitted as country:Jamaica when a validated candidate resolves to Jamaica: a country name after from/in/to/of/with/by, a curated demonym for Jamaica, or a curated foreign city mapped to Jamaica.

```

127:     "Jamaica",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.22. country:Russia (25 (0.2%))

Rule: Emitted as country:Russia when a validated candidate resolves to Russia: a country name after from/in/to/of/with/by, a curated demonym for Russia, or a curated foreign city mapped to Russia.

```

182:      "Russia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.23. country:Nicaragua (22 (0.2%))

Rule: Emitted as country:Nicaragua when a validated candidate resolves to Nicaragua: a country name after from/in/to/of/with/by, a curated demonym for Nicaragua, or a curated foreign city mapped to Nicaragua.


```

167:     "Nicaragua",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.24. country:Pakistan (15 (0.1%))

Rule: Emitted as country:Pakistan when a validated candidate resolves to Pakistan: a country name after from/in/to/of/with/by, a curated demonym for Pakistan, or a curated foreign city mapped to Pakistan.

```

172:     "Pakistan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.25. country:Ukraine (15 (0.1%))

Rule: Emitted as country:Ukraine when a validated candidate resolves to Ukraine: a country name after from/in/to/of/with/by, a curated demonym for Ukraine, or a curated foreign city mapped to Ukraine.

```

215:      "Ukraine",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.26. country:Georgia (14 (0.1%))

Rule: Emitted as country:Georgia when a validated candidate resolves to Georgia: a country name after from/in/to/of/with/by, a curated demonym for Georgia, or a curated foreign city mapped to Georgia.

```

108:      "Georgia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.27. country:Nigeria (13 (0.1%))

Rule: Emitted as country:Nigeria when a validated candidate resolves to Nigeria: a country name after from/in/to/of/with/by, a curated demonym for Nigeria, or a curated foreign city mapped to Nigeria.

```

169:      "Nigeria",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.28. country:Argentina (11 (0.1%))

Rule: Emitted as country:Argentina when a validated candidate resolves to Argentina: a country name after from/in/to/of/with/by, a curated demonym for Argentina, or a curated foreign city mapped to Argentina.

```

55:      "Argentina",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.29. country:Peru (11 (0.1%))

Rule: Emitted as country:Peru when a validated candidate resolves to Peru: a country name after from/in/to/of/with/by, a curated demonym for Peru, or a curated foreign city mapped to Peru.

```

176:      "Peru",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683: # country:<NAME> - broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:     country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.30. country:Lebanon (10 (0.1%))

Rule: Emitted as country:Lebanon when a validated candidate resolves to Lebanon: a country name after from/in/to/of/with/by, a curated demonym for Lebanon, or a curated foreign city mapped to Lebanon.

```

138:      "Lebanon",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.31. country:United-States (10 (0.1%))

Rule: Emitted as country:United-States when a validated candidate resolves to United States: a country name after from/in/to/of/with/by, a curated demonym for United States, or a curated foreign city mapped to United States.


```

220:     "United States",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.32. country:Egypt (8 (0.1%))

Rule: Emitted as country:Egypt when a validated candidate resolves to Egypt: a country name after from/in/to/of/with/by, a curated demonym for Egypt, or a curated foreign city mapped to Egypt.

```

97:      "Egypt",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.33. country:Iraq (8 (0.1%))

Rule: Emitted as country:Iraq when a validated candidate resolves to Iraq: a country name after from/in/to/of/with/by, a curated demonym for Iraq, or a curated foreign city mapped to Iraq.

```

123:      "Iraq",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*," )
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.34. country:Panama (8 (0.1%))

Rule: Emitted as country:Panama when a validated candidate resolves to Panama: a country name after from/in/to/of/with/by, a curated demonym for Panama, or a curated foreign city mapped to Panama.

```

174:     "Panama",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `*_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.35. country:Cambodia (7 (0.1%))

Rule: Emitted as country:Cambodia when a validated candidate resolves to Cambodia: a country name after from/in/to/of/with/by, a curated demonym for Cambodia, or a curated foreign city mapped to Cambodia.

```

77:     "Cambodia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.36. country:Sierra-Leone (7 (0.1%))

Rule: Emitted as country:Sierra-Leone when a validated candidate resolves to Sierra Leone: a country name after from/in/to/of/with/by, a curated demonym for Sierra Leone, or a curated foreign city mapped to Sierra Leone.

```

188:     "Sierra Leone",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}}\s+({_PLACE}))\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.37. country:South-Korea (7 (0.1%))

Rule: Emitted as country:South-Korea when a validated candidate resolves to South Korea: a country name after from/in/to/of/with/by, a curated demonym for South Korea, or a curated foreign city mapped to South Korea.

```

194:     "South Korea",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.38. country:France (6 (0.1%))

Rule: Emitted as country:France when a validated candidate resolves to France: a country name after from/in/to/of/with/by, a curated demonym for France, or a curated foreign city mapped to France.

```

105:      "France",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.39. country:Guyana (6 (0.1%))

Rule: Emitted as country:Guyana when a validated candidate resolves to Guyana: a country name after from/in/to/of/with/by, a curated demonym for Guyana, or a curated foreign city mapped to Guyana.


```

115:      "Guyana",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.40. country:Jordan (6 (0.1%))

Rule: Emitted as country:Jordan when a validated candidate resolves to Jordan: a country name after from/in/to/of/with/by, a curated demonym for Jordan, or a curated foreign city mapped to Jordan.

```

129:     "Jordan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.41. country:Sudan (6 (0.1%))

Rule: Emitted as country:Sudan when a validated candidate resolves to Sudan: a country name after from/in/to/of/with/by, a curated demonym for Sudan, or a curated foreign city mapped to Sudan.

```

198:      "Sudan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.42. country:Syria (6 (0.1%))

Rule: Emitted as country:Syria when a validated candidate resolves to Syria: a country name after from/in/to/of/with/by, a curated demonym for Syria, or a curated foreign city mapped to Syria.

```

202:     "Syria",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.43. country:Ghana (5 (0.0%))

Rule: Emitted as country:Ghana when a validated candidate resolves to Ghana: a country name after from/in/to/of/with/by, a curated demonym for Ghana, or a curated foreign city mapped to Ghana.

```

110:      "Ghana",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.44. country:Japan (5 (0.0%))

Rule: Emitted as country:Japan when a validated candidate resolves to Japan: a country name after from/in/to/of/with/by, a curated demonym for Japan, or a curated foreign city mapped to Japan.

```

128:     "Japan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.45. country:Myanmar (5 (0.0%))

Rule: Emitted as country:Myanmar when a validated candidate resolves to Myanmar: a country name after from/in/to/of/with/by, a curated demonym for Myanmar, or a curated foreign city mapped to Myanmar.

```

160:      "Myanmar",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*," )
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}}\s+({_PLACE}))\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.46. country:Turkey (5 (0.0%))

Rule: Emitted as country:Turkey when a validated candidate resolves to Turkey: a country name after from/in/to/of/with/by, a curated demonym for Turkey, or a curated foreign city mapped to Turkey.

```

212:     "Turkey",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.47. country:Barbados (4 (0.0%))

Rule: Emitted as country:Barbados when a validated candidate resolves to Barbados: a country name after from/in/to/of/with/by, a curated demonym for Barbados, or a curated foreign city mapped to Barbados.


```

63:      "Barbados",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.48. country:Cameroon (4 (0.0%))

Rule: Emitted as country:Cameroon when a validated candidate resolves to Cameroon: a country name after from/in/to/of/with/by, a curated demonym for Cameroon, or a curated foreign city mapped to Cameroon.

```

78:      "Cameroon",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.49. country:Chile (4 (0.0%))

Rule: Emitted as country:Chile when a validated candidate resolves to Chile: a country name after from/in/to/of/with/by, a curated demonym for Chile, or a curated foreign city mapped to Chile.

```

81:     "Chile",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `*_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.50. country:Ethiopia (4 (0.0%))

Rule: Emitted as country:Ethiopia when a validated candidate resolves to Ethiopia: a country name after from/in/to/of/with/by, a curated demonym for Ethiopia, or a curated foreign city mapped to Ethiopia.

```

102:     "Ethiopia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.51. country:Germany (4 (0.0%))

Rule: Emitted as country:Germany when a validated candidate resolves to Germany: a country name after from/in/to/of/with/by, a curated demonym for Germany, or a curated foreign city mapped to Germany.

```

109:      "Germany",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.52. country:Greece (4 (0.0%))

Rule: Emitted as country:Greece when a validated candidate resolves to Greece: a country name after from/in/to/of/with/by, a curated demonym for Greece, or a curated foreign city mapped to Greece.

```

111:      "Greece",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.53. country:Ireland (4 (0.0%))

Rule: Emitted as country:Ireland when a validated candidate resolves to Ireland: a country name after from/in/to/of/with/by, a curated demonym for Ireland, or a curated foreign city mapped to Ireland.

```

124:      "Ireland",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*(?)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.54. country:Kenya (4 (0.0%))

Rule: Emitted as country:Kenya when a validated candidate resolves to Kenya: a country name after from/in/to/of/with/by, a curated demonym for Kenya, or a curated foreign city mapped to Kenya.

```

131:      "Kenya",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.55. country:Liberia (4 (0.0%))

Rule: Emitted as country:Liberia when a validated candidate resolves to Liberia: a country name after from/in/to/of/with/by, a curated demonym for Liberia, or a curated foreign city mapped to Liberia.


```

140:     "Liberia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.56. country:Philippines (4 (0.0%))

Rule: Emitted as country:Philippines when a validated candidate resolves to Philippines: a country name after from/in/to/of/with/by, a curated demonym for Philippines, or a curated foreign city mapped to Philippines.

```

177:     "Philippines",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.57. country:Qatar (4 (0.0%))

Rule: Emitted as country:Qatar when a validated candidate resolves to Qatar: a country name after from/in/to/of/with/by, a curated demonym for Qatar, or a curated foreign city mapped to Qatar.

```

180:     "Qatar",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.58. country:Rwanda (4 (0.0%))

Rule: Emitted as country:Rwanda when a validated candidate resolves to Rwanda: a country name after from/in/to/of/with/by, a curated demonym for Rwanda, or a curated foreign city mapped to Rwanda.

```

183:      "Rwanda",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.59. country:South-Africa (4 (0.0%))

Rule: Emitted as country:South-Africa when a validated candidate resolves to South Africa: a country name after from/in/to/of/with/by, a curated demonym for South Africa, or a curated foreign city mapped to South Africa.

```

193:     "South Africa",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}}\s+({_PLACE}))\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.60. country:Trinidad (4 (0.0%))

Rule: Emitted as country:Trinidad when a validated candidate resolves to Trinidad: a country name after from/in/to/of/with/by, a curated demonym for Trinidad, or a curated foreign city mapped to Trinidad.

```

209:     "Trinidad",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `*_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({:_PLACE_})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({:_PLACE_})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({:_PLACE_})\s+({:_PLACE_})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.61. country:Australia (3 (0.0%))

Rule: Emitted as country:Australia when a validated candidate resolves to Australia: a country name after from/in/to/of/with/by, a curated demonym for Australia, or a curated foreign city mapped to Australia.

```

57:      "Australia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.62. country:Bahrain (3 (0.0%))

Rule: Emitted as country:Bahrain when a validated candidate resolves to Bahrain: a country name after from/in/to/of/with/by, a curated demonym for Bahrain, or a curated foreign city mapped to Bahrain.

```

61:      "Bahrain",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.63. country:Bangladesh (3 (0.0%))

Rule: Emitted as country:Bangladesh when a validated candidate resolves to Bangladesh: a country name after from/in/to/of/with/by, a curated demonym for Bangladesh, or a curated foreign city mapped to Bangladesh.


```

62:      "Bangladesh",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.64. country:Costa-Rica (3 (0.0%))

Rule: Emitted as country:Costa-Rica when a validated candidate resolves to Costa Rica: a country name after from/in/to/of/with/by, a curated demonym for Costa Rica, or a curated foreign city mapped to Costa Rica.

```

86:      "Costa Rica",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.65. country:Italy (3 (0.0%))

Rule: Emitted as country:Italy when a validated candidate resolves to Italy: a country name after from/in/to/of/with/by, a curated demonym for Italy, or a curated foreign city mapped to Italy.

```

126:     "Italy",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.66. country:Senegal (3 (0.0%))

Rule: Emitted as country:Senegal when a validated candidate resolves to Senegal: a country name after from/in/to/of/with/by, a curated demonym for Senegal, or a curated foreign city mapped to Senegal.

```

185:     "Senegal",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.67. country:Spain (3 (0.0%))

Rule: Emitted as country:Spain when a validated candidate resolves to Spain: a country name after from/in/to/of/with/by, a curated demonym for Spain, or a curated foreign city mapped to Spain.

```

196:     "Spain",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.68. country:Thailand (3 (0.0%))

Rule: Emitted as country:Thailand when a validated candidate resolves to Thailand: a country name after from/in/to/of/with/by, a curated demonym for Thailand, or a curated foreign city mapped to Thailand.

```

206:      "Thailand",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683: # country:<NAME> - broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:     country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:     if country and country.lower() in COUNTRY_LOWER:
1701:         add(f"country:{_normalize_country(country)}", span=m.span(1))

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     ""Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.69. country:Uganda (3 (0.0%))

Rule: Emitted as country:Uganda when a validated candidate resolves to Uganda: a country name after from/in/to/of/with/by, a curated demonym for Uganda, or a curated foreign city mapped to Uganda.

```

214:      "Uganda",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683: # country:<NAME> - broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:     country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:     if country and country.lower() in COUNTRY_LOWER:
1701:         add(f"country:{_normalize_country(country)}", span=m.span(1))

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.70. country:Uzbekistan (3 (0.0%))

Rule: Emitted as country:Uzbekistan when a validated candidate resolves to Uzbekistan: a country name after from/in/to/of/with/by, a curated demonym for Uzbekistan, or a curated foreign city mapped to Uzbekistan.

```

223:     "Uzbekistan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.71. country:Angola (2 (0.0%))

Rule: Emitted as country:Angola when a validated candidate resolves to Angola: a country name after from/in/to/of/with/by, a curated demonym for Angola, or a curated foreign city mapped to Angola.


```

54:      "Angola",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.72. country:Armenia (2 (0.0%))

Rule: Emitted as country:Armenia when a validated candidate resolves to Armenia: a country name after from/in/to/of/with/by, a curated demonym for Armenia, or a curated foreign city mapped to Armenia.

```

56:      "Armenia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `*_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.73. country:Azerbaijan (2 (0.0%))

Rule: Emitted as country:Azerbaijan when a validated candidate resolves to Azerbaijan: a country name after from/in/to/of/with/by, a curated demonym for Azerbaijan, or a curated foreign city mapped to Azerbaijan.

```

59:      "Azerbaijan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE_}\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.74. country:Burkina-Faso (2 (0.0%))

Rule: Emitted as country:Burkina-Faso when a validated candidate resolves to Burkina Faso: a country name after from/in/to/of/with/by, a curated demonym for Burkina Faso, or a curated foreign city mapped to Burkina Faso.

```

75:      "Burkina Faso",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.75. country:Finland (2 (0.0%))

Rule: Emitted as country:Finland when a validated candidate resolves to Finland: a country name after from/in/to/of/with/by, a curated demonym for Finland, or a curated foreign city mapped to Finland.

```

104:      "Finland",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683: # country:<NAME> - broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:     country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.76. country:Gabon (2 (0.0%))

Rule: Emitted as country:Gabon when a validated candidate resolves to Gabon: a country name after from/in/to/of/with/by, a curated demonym for Gabon, or a curated foreign city mapped to Gabon.

```

106:      "Gabon",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.77. country:Kazakhstan (2 (0.0%))

Rule: Emitted as country:Kazakhstan when a validated candidate resolves to Kazakhstan: a country name after from/in/to/of/with/by, a curated demonym for Kazakhstan, or a curated foreign city mapped to Kazakhstan.

```

130:      "Kazakhstan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.78. country:Kyrgyzstan (2 (0.0%))

Rule: Emitted as country:Kyrgyzstan when a validated candidate resolves to Kyrgyzstan: a country name after from/in/to/of/with/by, a curated demonym for Kyrgyzstan, or a curated foreign city mapped to Kyrgyzstan.

```

135:     "Kyrgyzstan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.79. country:Portugal (2 (0.0%))

Rule: Emitted as country:Portugal when a validated candidate resolves to Portugal: a country name after from/in/to/of/with/by, a curated demonym for Portugal, or a curated foreign city mapped to Portugal.


```

179:     "Portugal",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.80. country:Romania (2 (0.0%))

Rule: Emitted as country:Romania when a validated candidate resolves to Romania: a country name after from/in/to/of/with/by, a curated demonym for Romania, or a curated foreign city mapped to Romania.

```

181:      "Romania",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.81. country:South-Sudan (2 (0.0%))

Rule: Emitted as country:South-Sudan when a validated candidate resolves to South Sudan: a country name after from/in/to/of/with/by, a curated demonym for South Sudan, or a curated foreign city mapped to South Sudan.

```

195:     "South Sudan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.82. country:United-Kingdom (2 (0.0%))

Rule: Emitted as country:United-Kingdom when a validated candidate resolves to United Kingdom: a country name after from/in/to/of/with/by, a curated demonym for United Kingdom, or a curated foreign city mapped to United Kingdom.

```

218:     "United Kingdom",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.83. country:Uruguay (2 (0.0%))

Rule: Emitted as country:Uruguay when a validated candidate resolves to Uruguay: a country name after from/in/to/of/with/by, a curated demonym for Uruguay, or a curated foreign city mapped to Uruguay.

```

222:     "Uruguay",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.84. country:Yemen (2 (0.0%))

Rule: Emitted as country:Yemen when a validated candidate resolves to Yemen: a country name after from/in/to/of/with/by, a curated demonym for Yemen, or a curated foreign city mapped to Yemen.

```

227:     "Yemen",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE_}\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.85. country:Algeria (1 (0.0%))

Rule: Emitted as country:Algeria when a validated candidate resolves to Algeria: a country name after from/in/to/of/with/by, a curated demonym for Algeria, or a curated foreign city mapped to Algeria.

```

52:      "Algeria",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.86. country:Belarus (1 (0.0%))

Rule: Emitted as country:Belarus when a validated candidate resolves to Belarus: a country name after from/in/to/of/with/by, a curated demonym for Belarus, or a curated foreign city mapped to Belarus.

```

64:      "Belarus",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.87. country:Belgium (1 (0.0%))

Rule: Emitted as country:Belgium when a validated candidate resolves to Belgium: a country name after from/in/to/of/with/by, a curated demonym for Belgium, or a curated foreign city mapped to Belgium.


```

65:      "Belgium",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*(?)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({:_PLACE_})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({:_PLACE_})\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({:_PLACE_})\s+({:_PLACE_})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.88. country:Belize (1 (0.0%))

Rule: Emitted as country:Belize when a validated candidate resolves to Belize: a country name after from/in/to/of/with/by, a curated demonym for Belize, or a curated foreign city mapped to Belize.

```

66:      "Belize",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.89. country:Botswana (1 (0.0%))

Rule: Emitted as country:Botswana when a validated candidate resolves to Botswana: a country name after from/in/to/of/with/by, a curated demonym for Botswana, or a curated foreign city mapped to Botswana.

```

71:      "Botswana",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({:_PLACE_})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({:_PLACE_})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({:_PLACE_})\s+({:_PLACE_})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.90. country:Chad (1 (0.0%))

Rule: Emitted as country:Chad when a validated candidate resolves to Chad: a country name after from/in/to/of/with/by, a curated demonym for Chad, or a curated foreign city mapped to Chad.

```

80:      "Chad",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.91. country:Congo (1 (0.0%))

Rule: Emitted as country:Congo when a validated candidate resolves to Congo: a country name after from/in/to/of/with/by, a curated demonym for Congo, or a curated foreign city mapped to Congo.

```

85:      "Congo",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.92. country:Djibouti (1 (0.0%))

Rule: Emitted as country:Djibouti when a validated candidate resolves to Djibouti: a country name after from/in/to/of/with/by, a curated demonym for Djibouti, or a curated foreign city mapped to Djibouti.

```

93:      "Djibouti",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.93. country:Dominica (1 (0.0%))

Rule: Emitted as country:Dominica when a validated candidate resolves to Dominica: a country name after from/in/to/of/with/by, a curated demonym for Dominica, or a curated foreign city mapped to Dominica.

```

94:      "Dominica",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE_}\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """"Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.94. country:Iceland (1 (0.0%))

Rule: Emitted as country:Iceland when a validated candidate resolves to Iceland: a country name after from/in/to/of/with/by, a curated demonym for Iceland, or a curated foreign city mapped to Iceland.

```

119:      "Iceland",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.95. country:Latvia (1 (0.0%))

Rule: Emitted as country:Latvia when a validated candidate resolves to Latvia: a country name after from/in/to/of/with/by, a curated demonym for Latvia, or a curated foreign city mapped to Latvia.


```

137:     "Latvia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.96. country:Mauritania (1 (0.0%))

Rule: Emitted as country:Mauritania when a validated candidate resolves to Mauritania: a country name after from/in/to/of/with/by, a curated demonym for Mauritania, or a curated foreign city mapped to Mauritania.

```

151:     "Mauritania",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.97. country:Mauritius (1 (0.0%))

Rule: Emitted as country:Mauritius when a validated candidate resolves to Mauritius: a country name after from/in/to/of/with/by, a curated demonym for Mauritius, or a curated foreign city mapped to Mauritius.

```

152:      "Mauritius",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.98. country:Montenegro (1 (0.0%))

Rule: Emitted as country:Montenegro when a validated candidate resolves to Montenegro: a country name after from/in/to/of/with/by, a curated demonym for Montenegro, or a curated foreign city mapped to Montenegro.

```

157:      "Montenegro",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.99. country:Namibia (1 (0.0%))

Rule: Emitted as country:Namibia when a validated candidate resolves to Namibia: a country name after from/in/to/of/with/by, a curated demonym for Namibia, or a curated foreign city mapped to Namibia.

```

162:      "Namibia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.100. country:Poland (1 (0.0%))

Rule: Emitted as country:Poland when a validated candidate resolves to Poland: a country name after from/in/to/of/with/by, a curated demonym for Poland, or a curated foreign city mapped to Poland.

```

178:      "Poland",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.101. country:Togo (1 (0.0%))

Rule: Emitted as country:Togo when a validated candidate resolves to Togo: a country name after from/in/to/of/with/by, a curated demonym for Togo, or a curated foreign city mapped to Togo.

```

207:      "Togo",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `*_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:i:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:i:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:     # country:<NAME> - broader: any contextual country mention.
1684:     for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:         resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:         if resolved:
1687:             add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:     # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:     # country, which the bare-name validators above can't see. A demonym
1691:     # followed by a person noun also earns origin:.
1692:     for m in PATTERN_DEMONYM.finditer(text):
1693:         country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:         if country and country.lower() in COUNTRY_LOWER:
1695:             add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:             if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                 add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:     for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:         country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:         if country and country.lower() in COUNTRY_LOWER:
1701:             add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
2052:     if low == "burma":
2053:         return "Myanmar"
2054:     return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.102. country:Tonga (1 (0.0%))

Rule: Emitted as country:Tonga when a validated candidate resolves to Tonga: a country name after from/in/to/of/with/by, a curated demonym for Tonga, or a curated foreign city mapped to Tonga.

```

208:      "Tonga",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1683:      # country:<NAME> - broader: any contextual country mention.
1684:      for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:          resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:          if resolved:
1687:              add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689:      # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690:      # country, which the bare-name validators above can't see. A demonym
1691:      # followed by a person noun also earns origin:.
1692:      for m in PATTERN_DEMONYM.finditer(text):
1693:          country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:          if country and country.lower() in COUNTRY_LOWER:
1695:              add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:              if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:                  add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698:      for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:          country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

1700:          if country and country.lower() in COUNTRY_LOWER:
1701:              add(f"country:{_normalize_country(country)}", span=m.span(1))

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
2052:      if low == "burma":
2053:          return "Myanmar"
2054:      return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

3.103. country:United-Arab-Emirates (1 (0.0%))

Rule: Emitted as country:United-Arab-Emirates when a validated candidate resolves to United Arab Emirates: a country name after from/in/to/of/with/by, a curated demonym for United Arab Emirates, or a curated foreign city mapped to United Arab Emirates.


```

216:      "United Arab Emirates",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1683: # country:<NAME> - broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:     country = FOREIGN_CITY_TO_COUNTRY.get(m.group(1).lower())

```

```

1700:     if country and country.lower() in COUNTRY_LOWER:
1701:         add(f"country:{_normalize_country(country)}", span=m.span(1))

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:
2062:     """Resolve a candidate (any case, possibly two words) to its canonical
2063:     vocab spelling.
2064:
2065:     Tries the full phrase first, then its leading word, so a greedy
2066:     case-insensitive capture that swallowed a trailing lowercase token

```

4. crime

The crime namespace is produced by iterating CRIME_VOCAB and CRIME_SUBTYPE_VOCAB. Each regex match emits one crime tag. Some matches also emit parent tags through CRIME_PARENT_TAGS, such as rape -> sexual, murder -> homicide, fentanyl -> narcotics.

4.1. crime:assault (911 (9.1%))

Rule: Matches the assault entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

297: ("assault", r"\bassault(?:ed|ing|s)?\b"),

1638: # crime:<TYPE> - every distinct match emits one tag entry.
1639: for slug, pat_str in CRIME_VOCAB:
1640:     for m in re.finditer(pat_str, text, re.I):
1641:         add(f"crime:{slug}", span=m.span())
1642:         for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:             add(parent, span=m.span())
1644: for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:     for m in re.finditer(pat_str, text, re.I):
1646:         tag = f"crime:{slug}"
1647:         add(tag, span=m.span())
1648:         for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:             add(parent, span=m.span())
1650:
1651: if m := _martyrdom_match(text, account_category, entries):
1652:     add("theme:martyrdom", span=m.span())
1653:

```

4.2. crime:homicide (737 (7.4%))

Rule: Matches the homicide entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```
293:      ("homicide", r"\bhomicides?\b"),

1638:  # crime:<TYPE> – every distinct match emits one tag entry.
1639:  for slug, pat_str in CRIME_VOCAB:
1640:      for m in re.finditer(pat_str, text, re.I):
1641:          add(f"crime:{slug}", span=m.span())
1642:          for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:              add(parent, span=m.span())
1644:  for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:      for m in re.finditer(pat_str, text, re.I):
1646:          tag = f"crime:{slug}"
1647:          add(tag, span=m.span())
1648:          for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:              add(parent, span=m.span())
1650:
1651:  if m := _martyrdom_match(text, account_category, entries):
1652:      add("theme:martyrdom", span=m.span())
1653:
```

4.3. crime:sexual (619 (6.2%))

Rule: Matches the sexual entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```
305:      ("sexual", r"\b(?:sexual\s+assault|sex\s+crime|sex\s+offen[sc]e|sex\s+abuse)\b"),

333:      "crime:child-sexual": ("crime:sexual",),
334:      "crime:perjury": ("crime:disobedience",),

1638:  # crime:<TYPE> – every distinct match emits one tag entry.
1639:  for slug, pat_str in CRIME_VOCAB:
1640:      for m in re.finditer(pat_str, text, re.I):
1641:          add(f"crime:{slug}", span=m.span())
1642:          for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:              add(parent, span=m.span())
1644:  for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:      for m in re.finditer(pat_str, text, re.I):
1646:          tag = f"crime:{slug}"
1647:          add(tag, span=m.span())
1648:          for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:              add(parent, span=m.span())
1650:
1651:  if m := _martyrdom_match(text, account_category, entries):
1652:      add("theme:martyrdom", span=m.span())
1653:
```

4.4. crime:murder (614 (6.1%))

Rule: Matches the murder entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```
327: CRIME_SUBTYPE_VOCAB: tuple[tuple[str, str], ...] = (("murder", r"\bmurder(?:s|ed|ers?|ing)?\b"),)

1638:  # crime:<TYPE> – every distinct match emits one tag entry.
1639:  for slug, pat_str in CRIME_VOCAB:
1640:      for m in re.finditer(pat_str, text, re.I):
1641:          add(f"crime:{slug}", span=m.span())
1642:          for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:              add(parent, span=m.span())
1644:  for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:      for m in re.finditer(pat_str, text, re.I):
1646:          tag = f"crime:{slug}"
1647:          add(tag, span=m.span())
1648:          for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:              add(parent, span=m.span())
1650:
1651:  if m := _martyrdom_match(text, account_category, entries):
1652:      add("theme:martyrdom", span=m.span())
1653:
```

4.5. crime:narcotics (516 (5.2%))

Rule: Matches the narcotics entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

315:      ("narcotics", r"\bnarcotic[s]?\\b"),
335:      "crime:fentanyl": ("crime:narcotics",),
336:      "crime:cocaine": ("crime:narcotics",),

1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.6. crime:trafficking (405 (4.0%))

Rule: Matches the trafficking entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

302:      ("trafficking", r"\btraffick(?:ing|ers)\\b"),

1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.7. crime:weapon (377 (3.8%))

Rule: Matches the weapon entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

323:      ("weapon", r"\bweapon[s]?\\b|\\billegal firearm[s]?\\b"),

1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.8. crime:gang (337 (3.4%))

Rule: Matches the gang entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

312:      ("gang", r"\bgang (?:(member|members|affiliation)\b|\bgang-related\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.9. crime:rape (276 (2.8%))

Rule: Matches the rape entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

291:      ("rape", r"\brap(?:e|ed|ing|ist)\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.10. crime:child (256 (2.6%))

Rule: Matches the child entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

311:      ("child", r"\bchild (?:(abuse|sex|pornography|molest)|child sexual\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.11. crime:fraud (226 (2.3%))

Rule: Matches the fraud entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

316:      ("fraud", r"\bfraud(?:ulent|ster)?\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.12. crime:cocaine (222 (2.2%))

Rule: Matches the cocaine entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

300:      ("cocaine", r"\bcocaine\b"),
336:      "crime:cocaine": ("crime:narcotics",),
337:      "crime:meth": ("crime:narcotics",),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.13. crime:child-sexual (215 (2.1%))

Rule: Matches the child-sexual entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

333:      "crime:child-sexual": ("crime:sexual",),
334:      "crime:perjury": ("crime:disobedience",),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.14. crime:battery (187 (1.9%))

Rule: Matches the battery entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

298:      ("battery", r"\bbatter(?:y|ies|ed)\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.15. crime:firearm (177 (1.8%))

Rule: Matches the firearm entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

324:      ("firearm", r"\bfirearm[s]? \b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.16. crime:theft (166 (1.7%))

Rule: Matches the theft entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

295:      ("theft", r"\btheft\b|\bsteal(?:ing)?\b|\bstole\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.17. crime:robbery (164 (1.6%))

Rule: Matches the robbery entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

296:      ("robbery", r"\brobber(?:y|ies)\b|\brobbled\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.18. crime:fentanyl (156 (1.6%))

Rule: Matches the fentanyl entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

299:      ("fentanyl", r"\bfentanyl\b"),
335:      "crime:fentanyl": ("crime:narcotics",),
336:      "crime:cocaine": ("crime:narcotics",),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.19. crime:ms13 (156 (1.6%))

Rule: Matches the ms13 entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

313:      ("ms13", r"\bMS-?13\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.20. crime:dui (148 (1.5%))

Rule: Matches the dui entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

303:      ("dui", r"\bDUI\b|\bDWI\b"),

1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.21. crime:burglary (132 (1.3%))

Rule: Matches the burglary entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

294:      ("burglary", r"\bburglar(?:y|ize|ized|ies)\b"),

1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.22. crime:meth (132 (1.3%))

Rule: Matches the meth entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

301:      ("meth", r"\bmethamphetamine\b|\bmeth\b"),

337:      "crime:meth": ("crime:narcotics",),
338:  }

1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.23. crime:kidnap (123 (1.2%))

Rule: Matches the kidnap entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.


```

304:      ("kidnap", r"\bkidnap(?:ped|ping|per)?\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.24. crime:tren-de-aragua (117 (1.2%))

Rule: Matches the tren-de-aragua entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

314:      ("tren-de-aragua", r"\bTren de Aragua\b\bTdA\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.25. crime:arson (29 (0.3%))

Rule: Matches the arson entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

322:      ("arson", r"\barson(?:ist)?\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.26. crime:sodomy (23 (0.2%))

Rule: Matches the sodomy entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

292:      ("sodomy", r"\bsodom(?:y|ize|ized)\b"),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.27. crime:disobedience (8 (0.1%))

Rule: Matches the disobedience entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

334:      "crime:perjury": ("crime:disobedience",),
335:      "crime:fentanyl": ("crime:narcotics",),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

4.28. crime:perjury (1 (0.0%))

Rule: Matches the perjury entry in CRIME_VOCAB/CRIME_SUBTYPE_VOCAB. If this tag has a configured parent in CRIME_PARENT_TAGS, the parent is emitted from the same text span.

```

321:      ("perjury", r"\bperjur(?:y|ed)\b|\blying\s+under\s+oath\b"),
334:      "crime:perjury": ("crime:disobedience",),
335:      "crime:fentanyl": ("crime:narcotics",),
1638:      # crime:<TYPE> – every distinct match emits one tag entry.
1639:      for slug, pat_str in CRIME_VOCAB:
1640:          for m in re.finditer(pat_str, text, re.I):
1641:              add(f"crime:{slug}", span=m.span())
1642:              for parent in CRIME_PARENT_TAGS.get(f"crime:{slug}", ()):
1643:                  add(parent, span=m.span())
1644:      for slug, pat_str in CRIME_SUBTYPE_VOCAB:
1645:          for m in re.finditer(pat_str, text, re.I):
1646:              tag = f"crime:{slug}"
1647:              add(tag, span=m.span())
1648:              for parent in CRIME_PARENT_TAGS.get(tag, ()):
1649:                  add(parent, span=m.span())
1650:
1651:      if m := _martyrdom_match(text, account_category, entries):
1652:          add("theme:martyrdom", span=m.span())
1653:

```

5. event

The event namespace is produced by direct event regexes and disturbance-cluster regexes. Disturbance clusters require the city name and unrest/federal-response language within 180 characters of each other.

5.1. event:palestine (36 (0.4%))

Rule: Matches Palestine/Palestinian(s), Gaza/Gazan(s), Hamas, Israel-Hamas, Israel-Palestine/Palestinian, or West Bank.

```

639: PATTERN_EVENT_PALESTINE = _compile(
640:     r"\b(?:palestin(?:e|ian)s?|gaza(?:n)?s?|hamas|israel[- ]hamas|"
641:     r"israel[- ]palestin(?:e|ian)|west\s+bank)\b"
642: )
1551: (PATTERN_EVENT_PALESTINE, "event:palestine"),

```

5.2. event:los-angeles-disturbance (15 (0.1%))

Rule: Matches the los angeles disturbance disturbance cluster: the city name must occur within 180 characters of riot/unrest/protest/federal-response language, in either order.

```

731: (
732:     "los-angeles-disturbance",
733:     _compile(
734:         r"\b(?:Los\s+Angeles|L\.A\.|LA)\b.{0,180}\b(?:riot(?:s|ers|ing)?|"
735:         r"civil\s+disturbance|civil\s+unrest|violent\s+protests?|anti[- ]ICE\s+"
736:         r"?(?:riot(?:s|ers|ing)?|protests?)|federal\s+building|concrete\s+at\s+DHS\s+agents)\b"
737:         r"|\b(?:riot(?:s|ers|ing)?|civil\s+disturbance|civil\s+unrest|violent\s+protests?|"
738:         r"anti[- ]ICE\s+(?:riot(?:s|ers|ing)?|protests?)|federal\s+building|"
739:         r"concrete\s+at\s+DHS\s+agents)\b.{0,180}\b(?:Los\s+Angeles|L\.A\.|LA)\b"
740:     ),
1609:     if m := _coded_nativism_match(text, entries):
1610:         add("theme:nativism", span=m.span())
1611:
1612:     if m := _theme_religion_match(text):

```

5.3. event:portland-disturbance (6 (0.1%))

Rule: Matches the portland disturbance disturbance cluster: the city name must occur within 180 characters of riot/unrest/protest/federal-response language, in either order.

```

753: (
754:     "portland-disturbance",
755:     _compile(
756:         r"\bPortland\b.{0,180}\b(?:riot(?:s|ers|ing)?|civil\s+disturbance|civil\s+unrest|"
757:         r"violent\s+protests?|anti[- ]ICE\s+(?:riot(?:s|ers|ing)?|protests?)|"
758:         r"ICE\s+(?:detention|facility)|Antifa|domestic\s+terrorism)\b"
759:         r"|\b(?:riot(?:s|ers|ing)?|civil\s+disturbance|civil\s+unrest|violent\s+protests?|"
760:         r"anti[- ]ICE\s+(?:riot(?:s|ers|ing)?|protests?)|ICE\s+(?:detention|facility)|"
761:         r"Antifa|domestic\s+terrorism)\b.{0,180}\bPortland\b"
762:     ),
1609:     if m := _coded_nativism_match(text, entries):
1610:         add("theme:nativism", span=m.span())
1611:
1612:     if m := _theme_religion_match(text):

```

5.4. event:minneapolis-disturbance (5 (0.0%))

Rule: Matches the minneapolis disturbance disturbance cluster: the city name must occur within 180 characters of riot/unrest/protest/federal-response language, in either order.

```

742: (
743:     "minneapolis-disturbance",
744:     _compile(
745:         r"\bMinneapolis\b.{0,180}\b(?:Bovino|riot(?:s|ers|ing)?|civil\s+disturbance|"
746:         r"civil\s+unrest|violent\s+protests?|protests?|tear\s+gas|pellet\s+guns?|"
747:         r"heavy[- ]handed|lawsuits?|killing|killed|disaster)\b"
748:         r"|\b(?:Bovino|riot(?:s|ers|ing)?|civil\s+disturbance|civil\s+unrest|"
749:         r"violent\s+protests?|protests?|tear\s+gas|pellet\s+guns?|heavy[- ]handed|"
750:         r"lawsuits?|killing|killed|disaster)\b.{0,180}\bMinneapolis\b"
751:     ),
1609:     if m := _coded_nativism_match(text, entries):
1610:         add("theme:nativism", span=m.span())
1611:
1612:     if m := _theme_religion_match(text):

```

6. genre

The genre namespace is produced from produced-video regexes, parody franchise matches, media-description sidecar tags, and the lineup composite. The lineup composite is exact: reply + theme:criminal + exactly one attached media item.

6.1. genre:lineup (670 (6.7%))

Rule: Composite rule: tweet_type is reply, theme:criminal already fired, and media_count equals exactly 1.

```

1709: # genre:lineup: composite replies that hit theme:criminal with one photo.
1710: if (
1711:     tweet_type == "reply"
1712:     and any(e["tag"] == "theme:criminal" for e in entries)
1713:     and media_count == 1
1714: ):
1715:     add("genre:lineup")

```

6.2. genre:advertisement (439 (4.4%))

Rule: On video posts or sidecar text, matches new ad/advert/spot/commercial, ad/advertisement/commercial/promo/promotional video/campaign spot, campaign/recruitment ad/spot/video/campaign, join ICE/CBP/HSI language, join.ice.gov, apply today/now, Promises Made/Kept, or most secure border in American history.

```

1045: PATTERN_VIDEO_AD = _compile(
1046:     r"\b(?:new\s+(?:ad|advert|spot|commercial))\b"
1047:     r"|\b(?:ad|advertisement|commercial|promo|promotional\s+video|campaign\s+spot)\b"
1048:     r"|\b(?:campaign|recruitment)\s+(?:ad|spot|video|campaign)\b"
1049:     r"|\bjoin\s+(?:ice|cbp|hsi|the\s+(?:ice|cbp)\s+(?:team|family))\b"
1050:     r"|\bjoin\.ice\.gov\b"
1051:     r"|\bapply\s+(?:today|now)\b"
1052:     r"|\bpromises?\s+made[,.]?\s+promises?\s+kept\b"
1053:     r"|\bmost\s+secure\s+border\s+in\s+american\s+history\b"
1054: )

```

6.3. genre:psa (128 (1.3%))

Rule: On video posts or sidecar text, matches PSA/public service announcement, learn/find out/more info/visit website at, did you know, know your rights/facts/signs, protect yourself, stay safe, report a/suspicious, or call 1-800/888/877/866 numbers.

```

1019: PATTERN_VIDEO_PSA = _compile(
1020:     r"\b(?:psa|public\s+service\s+announcement)\b"
1021:     r"|\b(?:learn\s+more|find\s+out\s+more|more\s+info|visit\s+(?:our\s+)?website)\s+at\b"
1022:     r"|\b(?:did\s+you\s+know(?:\s+that)?\b"
1023:     r"|\b(?:know\s+(?:your\s+rights|the\s+facts|the\s+signs)|protect\s+yourself|stay\s+safe)"
1024:     r"report\s+(?:a|suspicious))\b"
1025:     r"|\bcall\s+(?:1-?800|1-?888|1-?877|1-?866)[- ]?\d"
1026: )

```

6.4. genre:recruitment (126 (1.3%))

Rule: On video posts or sidecar text, matches join.ice.gov, DHS/ICE/CBP/HSI career/join/job/recruiting URLs, hiring/recruiting, recruitment/hiring ad/spot/video/campaign, join ICE/CBP/HSI/DHS/team/family, apply today/now, answer the call, or serve your/our/their/his/her country/nation/community.

```

1033: PATTERN_VIDEO_RECRUITMENT = _compile(
1034:     r"\bjoin\.ice\.gov\b"
1035:     r"|\b(?:ice|cbp|hsi|dhs)\.gov/(?:careers?|join|jobs?|recruit(?:ment|ing)?|homeland-security-careers)\b"
1036:     r"|\b(?:careers?|jobs?)\.(?:ice|cbp|hsi|dhs)\.gov\b"
1037:     r"|\b(?:dhs|ice|cbp|hsi)\s+(?:is\s+)?(?:hiring|recruiting)\b"
1038:     r"|\b(?:recruitment|hiring)\s+(?:ad|spot|video|campaign)\b"
1039:     r"|\bjoin\s+(?:ice|cbp|hsi|dhs|the\s+(?:ice|cbp|hsi|dhs)\s+(?:team|family))\b"
1040:     r"|\bapply\s+(?:today|now)\b"
1041:     r"|\bcareer(?:s)?\s+(?:with|at)\s+(?:ice|cbp|hsi|dhs)\b"
1042:     r"|\banswer\s+the\s+call\b"
1043:     r"|\bserve\s+(?:your|our|their|his|her)\s+(?:country|nation|countrymen|community)\b"
1044: )

```

6.5. genre:utopian (44 (0.4%))

Rule: Matches utopian, idealized, aspirational, golden age, bright future, morning in America, or sunlit/heroic/triumphal within 80 characters of montage/vision/future/aesthetic.

```

1067:     ),
1068:     (
1069:         "genre:utopian",
1070:         _compile(
1071:             r"\butopian\b|\bidealized\b|\baspirational\b|\b(?:golden\s+age|bright\s+future|morning\s+in\s+america)\b"
1072:             r"|\b(?:sunlit|heroic|triumphal)\b.{0,80}\b(?:montage|vision|future|aesthetic)\b"
1073:         ),
1074:     ),
1075:     (
1076:         "genre:dystopian",
1077:         _compile(
1078:             r"\bdystopian\b|\bsci[- ]?fi\b|\bscience[- ]fiction\b|\bcyberpunk\b"
1079:             r"|\b(?:dark|bleak|apocalyptic|hellscape|surveillance[- ]state|futuristic)\b.{0,80}\b(?:future|city|vision|scene|aesthetic)\b"
1080:         ),
1081:     ),
1082: )
1083:
1084: # long : > 120s
1085: # We tag the bucket regardless of whether a kind matched so the viewer
1086: # can filter "all videos under 30s" without depending on text cues.
1087: if video_count > 0:
1088:     for pat, tag in (
1089:         (PATTERN_VIDEO_BODYCAM, "video:bodycam"),
1090:         (PATTERN_VIDEO_INTERVIEW, "video:interview"),
1091:         (PATTERN_VIDEO_NEWS_CLIP, "video:news-clip"),
1092:         (PATTERN_VIDEO_SPEECH, "video:speech"),
1093:     ):
1094:         m = pat.search(text)
1095:         if m:
1096:             add(tag, span=m.span())
1097:     if PATTERN_AUDIO_MUSIC_CONTEXT.search(text):
1098:         add("audio:music-likely")
1099:     if PATTERN_AUDIO_MUSIC_CONTEXT.search(reply_context_text):
1100:         add("audio:music-likely", source="reply-context")
1101:     if any(e["tag"] in {"genre:music-video"} for e in entries):
1102:         add("audio:music-likely")
1103:     if m := PATTERN_PRODUCED_VIDEO_STYLE.search(text):
1104:         add("video:produced", span=m.span())
1105:     if video_max_duration_sec is not None and video_max_duration_sec > 0:
1106:         if video_max_duration_sec <= 30:
1107:             add("video:short")
1108:         elif video_max_duration_sec <= 120:
1109:             add("video:medium")
1110:         else:
1111:             add("video:long")
1112:     # Derive genre:music-video ONLY from an explicit genre:music-video
1113:     # signal (set by the conservative describe_media / manual-review rules) –
1114:     # NEVER from audio:music-likely, which is an incidental acoustic heuristic.
1115:     # Also suppress it on speech / press-conference clips.
1116:     if not speech_indicator_present and any(e["tag"] == "genre:music-video" for e in entries):
1117:         add("genre:music-video")
1118:     if any(
1119:         e["tag"] in PRODUCED_VIDEO_STRUCTURE_TAGS or str(e["tag"]).startswith("genre:")
1120:         for e in entries
1121:     ):
1122:         add("video:produced")

```

6.6. genre:music-video (36 (0.4%))

Rule: On video posts, matches explicit music-video wording: official music video, set to music/the song/the track, official video/audio for, or lyric/lyrics video. Speech/press-conference indicators suppress this tag.

```

987: PATTERN_VIDEO_MUSIC_VIDEO = _compile(
988:     r"\b(?:official\s+)?music\s+video\b"
989:     r"|\bset\s+to\s+(?:music|the\s+song|the\s+track)\b"
990:     r"|\bofficial\s+(?:video|audio)\s+for\b"
991:     r"|\b(?:lyric|lyrics)\s+video\b"
992: )

```

6.7. genre:dystopian (18 (0.2%))

Rule: Matches dystopian, sci-fi/science fiction, cyberpunk, or dark/bleak/apocalyptic/hellscape/surveillance-state/futuristic within 80 characters of future/city/vision/scene/aesthetic.

```

1074:     ),
1075:     (
1076:         "genre:dystopian",
1077:         _compile(
1078:             r"\bdystopian\b|\bsci[- ]?fi\b|\bscience[- ]fiction\b|\bcyberpunk\b"
1079:             r"|\b(?:dark|bleak|apocalyptic|hellscape|surveillance[- ]state|futuristic)\b.{0,80}\b(?:future|city|vision|scene|aesthetic)\b"
1080:         ),
1081:     ),
1082: )
1083: PRODUCED_VIDEO_STRUCTURE_TAGS = {
1084:     "video:produced",
1085:     "genre:music-video",
1086:     "video:montage",
1087: }
1088:
1089: # long : > 120s
1090: # We tag the bucket regardless of whether a kind matched so the viewer
1091: # can filter "all videos under 30s" without depending on text cues.
1092: if video_count > 0:
1093:     for pat, tag in (
1094:         (PATTERN_VIDEO_BODYCAM, "video:bodycam"),
1095:         (PATTERN_VIDEO_INTERVIEW, "video:interview"),
1096:         (PATTERN_VIDEO_NEWS_CLIP, "video:news-clip"),
1097:         (PATTERN_VIDEO_SPEECH, "video:speech"),
1098:     ):
1099:         m = pat.search(text)
1100:         if m:
1101:             add(tag, span=m.span())
1102:     if PATTERN_AUDIO_MUSIC_CONTEXT.search(text):
1103:         add("audio:music-likely")
1104:     if PATTERN_AUDIO_MUSIC_CONTEXT.search(reply_context_text):
1105:         add("audio:music-likely", source="reply-context")
1106:     if any(e["tag"] in {"genre:music-video"} for e in entries):
1107:         add("audio:music-likely")
1108:     if m := PATTERN_PRODUCED_VIDEO_STYLE.search(text):
1109:         add("video:produced", span=m.span())
1110:         if video_max_duration_sec is not None and video_max_duration_sec > 0:
1111:             if video_max_duration_sec <= 30:
1112:                 add("video:short")
1113:             elif video_max_duration_sec <= 120:
1114:                 add("video:medium")
1115:             else:
1116:                 add("video:long")
1117:         # Derive genre:music-video ONLY from an explicit genre:music-video
1118:         # signal (set by the conservative describe_media / manual-review rules) –
1119:         # NEVER from audio:music-likely, which is an incidental acoustic heuristic.
1120:         # Also suppress it on speech / press-conference clips.
1121:         if not speech_indicator_present and any(e["tag"] == "genre:music-video" for e in entries):
1122:             add("genre:music-video")
1123:         if any(
1124:             e["tag"] in PRODUCED_VIDEO_STRUCTURE_TAGS or str(e["tag"]).startswith("genre:")
1125:             for e in entries
1126:         ):
1127:             add("video:produced")

```

6.8. genre:parody (5 (0.0%))

Rule: Emitted when any franchise parody pattern fires; it is not emitted by itself.

```

1656:     if m := franchise_pat.search(text):
1657:         add("genre:parody", span=m.span())
1658:         add(f"parody:{franchise_slug}", span=m.span())
1659:         add("theme:pop-culture-reference")
1660:     # Also catch "this is the way" + one more Star Wars cue (e.g. "a galaxy").
1661:     if (
1662:         PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1663:         and PATTERN_PARODY_STAR_WARS_EXTRA_CUE.search(text)
1664:         and not any(e["tag"] == "parody:star-wars" for e in entries)
1665:     ):
1666:         m_way = PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1667:         if m_way:
1668:             add("genre:parody", span=m_way.span())
1669:             add("parody:star-wars", span=m_way.span())
1670:             add("theme:pop-culture-reference")

```

6.9. genre:war-movie (2 (0.0%))

Rule: Matches war movie/war film/action movie, or cinematic/movie-trailer/trailer-style/dramatic within 80 characters of war/battle/combat/military/raid/operation in either order.

```

1059:     ("genre:advertisement", PATTERN_VIDEO_AD),
1060:     (
1061:         "genre:war-movie",
1062:         _compile(
1063:             r"\bwar[- ]movie\b\bwar[- ]film\b\baction[- ]movie\b"
1064:             r"|\b(?:cinematic|movie[- ]trailer|trailer[- ]style|dramatic)\b.{0,80}\b(?:war|battle|combat|military|raid|operation)\b"
1065:             r"|\b(?:war|battle|combat|military|raid|operation)\b.{0,80}\b(?:cinematic|movie[- ]trailer|trailer[- ]style|dramatic)\b"
1066:         ),
1067:     ),
1068:     (
1069:         "genre:utopian",
1070:         _compile(
1071:             r"\butopian\b\bidealized\b\baspirational\b\b(?:golden\s+age|bright\s+future|morning\s+in\s+america)\b"

1730: # long : > 120s
1731: # We tag the bucket regardless of whether a kind matched so the viewer
1732: # can filter "all videos under 30s" without depending on text cues.
1733: if video_count > 0:
1734:     for pat, tag in (
1735:         (PATTERN_VIDEO_BODYCAM, "video:bodycam"),
1736:         (PATTERN_VIDEO_INTERVIEW, "video:interview"),
1737:         (PATTERN_VIDEO_NEWS_CLIP, "video:news-clip"),
1738:         (PATTERN_VIDEO_SPEECH, "video:speech"),
1739:     ):
1740:         m = pat.search(text)
1741:         if m:
1742:             add(tag, span=m.span())
1743:         if PATTERN_AUDIO_MUSIC_CONTEXT.search(text):
1744:             add("audio:music-likely")
1745:         if PATTERN_AUDIO_MUSIC_CONTEXT.search(reply_context_text):
1746:             add("audio:music-likely", source="reply-context")
1747:         if any(e["tag"] in {"genre:music-video"} for e in entries):
1748:             add("audio:music-likely")
1749:         if m := PATTERN_PRODUCED_VIDEO_STYLE.search(text):

1750:             add("video:produced", span=m.span())
1751:         if video_max_duration_sec is not None and video_max_duration_sec > 0:
1752:             if video_max_duration_sec <= 30:
1753:                 add("video:short")
1754:             elif video_max_duration_sec <= 120:
1755:                 add("video:medium")
1756:             else:
1757:                 add("video:long")
1758:         # Derive genre:music-video ONLY from an explicit genre:music-video
1759:         # signal (set by the conservative describe_media / manual-review rules) –
1760:         # NEVER from audio:music-likely, which is an incidental acoustic heuristic.
1761:         # Also suppress it on speech / press-conference clips.
1762:         if not speech_indicator_present and any(e["tag"] == "genre:music-video" for e in entries):
1763:             add("genre:music-video")
1764:         if any(
1765:             e["tag"] in PRODUCED_VIDEO_STRUCTURE_TAGS or str(e["tag"]).startswith("genre:")
1766:             for e in entries
1767:         ):
1768:             add("video:produced")

```

7. legal

The legal namespace is produced by regexes for criminal prosecution posture, civil litigation posture, and the literal phrase birthright citizenship.

7.1. legal:criminal-prosecution (1,856 (18.5%))

Rule: Matches prosecution words: prosecute/prosecuted/prosecution, indict/indictment, charged with, criminal complaint/charge/case/prosecution, arraign/arraignment, plead guilty/not guilty, convict/conviction, sentence/sentencing, felony charges, or misdemeanor charges.

```

1178: PATTERN_LEGAL_CRIMINAL_PROSECUTION = _compile(
1179:     r"\bprosecut(?:e|ed|ing|ion|ions)\b"
1180:     r"|\bindict(?:ed|ment|ments)?\b"
1181:     r"|\bcharged\s+with\b"
1182:     r"|\bcriminal\s+(?:complaint|charges?|case|prosecution)\b"
1183:     r"|\barraign(?:ed|ment)?\b"
1184:     r"|\bble(?:a|d|aded|ads)\s+(?:guilty|not\s+guilty)\b"
1185:     r"|\bconvict(?:ed|ion|ions)?\b"
1186:     r"|\bsentenc(?:ed|ing)\b"
1187:     r"|\bfelony\s+charges?\b"
1188:     r"|\bmisdemeanor\s+charges?\b"
1189: )

1546:     (PATTERN_LEGAL_CRIMINAL_PROSECUTION, "legal:criminal-prosecution"),

```

7.2. legal:civil-lawsuit (30 (0.3%))

Rule: Matches civil lawsuit/suit/action/case/complaint/litigation, lawsuit, filed a lawsuit/suit/civil action/complaint, sue/sued/sues/suing, injunction, temporary restraining order, TRO, settlement agreement, or consent decree.

```

1190: PATTERN_LEGAL_CIVIL_LAWSUIT = _compile(
1191:     r"\bcivil\s+(?:lawsuit|suit|action|case|complaint|litigation)\b"
1192:     r"|\blawsuits?\b"
1193:     r"|\bfiled\s+(?:a\s+)?(?:lawsuit|suit|civil\s+action|complaint)\b"
1194:     r"|\bsu(?:e|ed|es|ing)\b"
1195:     r"|\binjunction\b"
1196:     r"|\bttemporary\s+restraining\s+order\b"
1197:     r"|\btRO\b"
1198:     r"|\bsettlement\s+agreement\b"
1199:     r"|\bconsent\s+decree\b"
1200: )

1547: (PATTERN_LEGAL_CIVIL_LAWSUIT, "legal:civil-lawsuit"),

```

7.3. legal:birthright-citizenship (3 (0.0%))

Rule: Matches the literal phrase birthright citizenship.

```

814: PATTERN_LEGAL_BIRTHRIGHT_CITIZENSHIP = _compile(r"\bbirthright\s+citizenship\b")

1562: (PATTERN_LEGAL_BIRTHRIGHT_CITIZENSHIP, "legal:birthright-citizenship"),

```

8. military

The military namespace is produced from text regexes for service branches and from account-mention aliases. Any military branch tag also adds topic:military.

8.1. military:navy (171 (1.7%))

Rule: Matches navy branch terms in MILITARY_VOCAB or a mention alias mapped to military:navy; then implies topic:military.

```

579: (
580:     "navy",
581:     _compile(
582:         r"\b(?:u\.s\.s+)?navy|sailors?|naval\s+academy|USNA|midshipmen|"
583:         r"carrier\s+strike\s+group|aircraft\s+carrier|carrier\s+air\s+wing|"
584:         r"CVN\s*d+|USS\s+[A-Z][A-Za-z0-9-]+|USNS\s+[A-Z][A-Za-z0-9-]+\)\b"
585:     ),
586: ),
587: (
588:     "air-force",

1631:     for slug, pat in MILITARY_VOCAB:
1632:         for m in pat.finditer(text):
1633:             add(f"military:{slug}", span=m.span())
1634:
1635:     if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (

```

8.2. military:coast-guard (142 (1.4%))

Rule: Matches coast guard branch terms in MILITARY_VOCAB or a mention alias mapped to military:coast-guard; then implies topic:military.


```

593: (
594:     "coast-guard",
595:     _compile(
596:         r"\b(?:coast\s+guard|USCG|USCGA|coast\s+guard\s+academy|coasties?|coast\s+guards?m[ae]n)\b"
597:     ),
598: ),
599: ("national-guard", _compile(r"\b(?:national\s+guard|national\s+guards?m[ae]n)\b")),
600: )
601: MILITARY_MENTION_ALIASES: dict[str, str] = {
602:     "usarmynorth": "army",

1631:     for slug, pat in MILITARY_VOCAB:
1632:         for m in pat.finditer(text):
1633:             add(f"military:{slug}", span=m.span())
1634:
1635:     if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (

```

8.3. military:army (43 (0.4%))

Rule: Matches army branch terms in MILITARY_VOCAB or a mention alias mapped to military:army; then implies topic:military.

```

577: MILITARY_VOCAB: tuple[tuple[str, re.Pattern[str]], ...] = (
578:     ("army", _compile(r"\b(?:?u\.s\.s+)?army|soldiers?|west\s+point|USMA)\b")),
579:     (
580:         "navy",
581:         _compile(
582:             r"\b(?:?u\.s\.s+)?navy|sailors?|naval\s+academy|USNA|midshipmen|"
583:             r"carrier\s+strike\s+group|aircraft\s+carrier|carrier\s+air\s+wing|"
584:             r"CVN\s+d+|USS\s+[A-Z][A-Za-z0-9-]+|USNS\s+[A-Z][A-Za-z0-9-]+\)\b"
585:         ),
586:     ),

1631:     for slug, pat in MILITARY_VOCAB:
1632:         for m in pat.finditer(text):
1633:             add(f"military:{slug}", span=m.span())
1634:
1635:     if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (

```

8.4. military:national-guard (38 (0.4%))

Rule: Matches national guard branch terms in MILITARY_VOCAB or a mention alias mapped to military:national-guard; then implies topic:military.

```

598: ),
599: ("national-guard", _compile(r"\b(?:national\s+guard|national\s+guards?m[ae]n)\b")),
600: )
601: MILITARY_MENTION_ALIASES: dict[str, str] = {
602:     "usarmynorth": "army",
603:     "usnationalguard": "national-guard",
604:     "usguard": "national-guard",
605:     "usnavy": "navy",
606:     "usairforce": "air-force",
607:     "usspaceforce": "space-force",

1631:     for slug, pat in MILITARY_VOCAB:
1632:         for m in pat.finditer(text):
1633:             add(f"military:{slug}", span=m.span())
1634:
1635:     if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (

```

8.5. military:air-force (37 (0.4%))

Rule: Matches air force branch terms in MILITARY_VOCAB or a mention alias mapped to military:air-force; then implies topic:military.

```
587: (
588:     "air-force",
589:     _compile(r"\b(?:?:u\.s\.s+)?air\s+force|usaf|airm[ae]n|air\s+force\s+academy\b"),
590: ),
591: ("space-force", _compile(r"\b(?:?:u\.s\.s+)?space\s+force|ussf\b")),
592: ("marines", _compile(r"\b(?:marine\s+corps|u\.s\.s+marines?|usmc|marines)\b")),
593: (
594:     "coast-guard",
595:     _compile(
596:         r"\b(?:coast\s+guard|USCG|USCGA|coast\s+guard\s+academy|coasties?|coast\s+guards?m[ae]n)\b"
597:     ),
598: ),
599: ("national-guard", _compile(r"\b(?:national\s+guard|national\s+guards?m[ae]n)\b")),
600: )

1631: for slug, pat in MILITARY_VOCAB:
1632:     for m in pat.finditer(text):
1633:         add(f"military:{slug}", span=m.span())
1634:
1635: if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (
```

8.6. military:marines (15 (0.1%))

Rule: Matches marines branch terms in MILITARY_VOCAB or a mention alias mapped to military:marines; then implies topic:military.

```
591: ("space-force", _compile(r"\b(?:?:u\.s\.s+)?space\s+force|ussf\b")),
592: ("marines", _compile(r"\b(?:marine\s+corps|u\.s\.s+marines?|usmc|marines)\b")),
593: (
594:     "coast-guard",
595:     _compile(
596:         r"\b(?:coast\s+guard|USCG|USCGA|coast\s+guard\s+academy|coasties?|coast\s+guards?m[ae]n)\b"
597:     ),
598: ),
599: ("national-guard", _compile(r"\b(?:national\s+guard|national\s+guards?m[ae]n)\b")),
600: )

1631: for slug, pat in MILITARY_VOCAB:
1632:     for m in pat.finditer(text):
1633:         add(f"military:{slug}", span=m.span())
1634:
1635: if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (
```

8.7. military:space-force (5 (0.0%))

Rule: Matches space force branch terms in MILITARY_VOCAB or a mention alias mapped to military:space-force; then implies topic:military.

```

590: ),
591: ("space-force", _compile(r"\b(?:?u\.s\.s+)?space\s+force|ussf\b")),
592: ("marines", _compile(r"\b(?:marine\s+corps|u\.s\.s+marines?|usmc|marines)\b")),
593: (
594:     "coast-guard",
595:     _compile(
596:         r"\b(?:coast\s+guard|USCG|USCGA|coast\s+guard\s+academy|coasties?|coast\s+guards?m[ae]n)\b"
597:     ),
598: ),
599: ("national-guard", _compile(r"\b(?:national\s+guard|national\s+guards?m[ae]n)\b")),

1631: for slug, pat in MILITARY_VOCAB:
1632:     for m in pat.finditer(text):
1633:         add(f"military:{slug}", span=m.span())
1634:
1635: if _general_topic_score(text) >= 3:

1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (

```

9. origin

The origin namespace is stricter than country. It is produced when the text says from <country>, with a comma after the country, or when a curated demonym is immediately followed by a person-origin noun such as national, citizen, immigrant, migrant, man, woman, descent, or origin.

9.1. origin:Mexico (378 (3.8%))

Rule: Emitted as origin:Mexico when the text has from Mexico, with a comma after the country, or a demonym for Mexico followed immediately by a person-origin noun.

```

153:     "Mexico",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!({_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," - anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:({_PREPOSITIONS})\s+({_PLACE}))\b")
1229: # "<Place>," <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.2. origin:El-Salvador (94 (0.9%))

Rule: Emitted as origin:El-Salvador when the text has from El Salvador, with a comma after the country, or a demonym for El Salvador followed immediately by a person-origin noun.

```

98:     "El Salvador",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.3. origin:Guatemala (91 (0.9%))

Rule: Emitted as origin:Guatemala when the text has from Guatemala, with a comma after the country, or a demonym for Guatemala followed immediately by a person-origin noun.

```

113:     "Guatemala",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.4. origin:Honduras (86 (0.9%))

Rule: Emitted as origin:Honduras when the text has from Honduras, with a comma after the country, or a demonym for Honduras followed immediately by a person-origin noun.

```

117: "Honduras",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.5. origin:Venezuela (54 (0.5%))

Rule: Emitted as origin:Venezuela when the text has from Venezuela, with a comma after the country, or a demonym for Venezuela followed immediately by a person-origin noun.

```

225: "Venezuela",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.6. origin:Cuba (52 (0.5%))

Rule: Emitted as origin:Cuba when the text has from Cuba, with a comma after the country, or a demonym for Cuba followed immediately by a person-origin noun.

```

88: "Cuba",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.7. origin:Ecuador (25 (0.2%))

Rule: Emitted as origin:Ecuador when the text has from Ecuador, with a comma after the country, or a demonym for Ecuador followed immediately by a person-origin noun.

```

96: "Ecuador",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.8. origin:China (23 (0.2%))

Rule: Emitted as origin:China when the text has from China, with a comma after the country, or a demonym for China followed immediately by a person-origin noun.

```

82: "China",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.9. origin:Iran (23 (0.2%))

Rule: Emitted as origin:Iran when the text has from Iran, with a comma after the country, or a demonym for Iran followed immediately by a person-origin noun.

```

122:     "Iran",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.10. origin:Laos (23 (0.2%))

Rule: Emitted as origin:Laos when the text has from Laos, with a comma after the country, or a demonym for Laos followed immediately by a person-origin noun.

```

136: "Laos",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({:_PLACE_})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({:_PLACE_})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({:_PLACE_})\s+({:_PLACE_})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.11. origin:Colombia (22 (0.2%))

Rule: Emitted as origin:Colombia when the text has from Colombia, with a comma after the country, or a demonym for Colombia followed immediately by a person-origin noun.

```

83:     "Colombia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.12. origin:Vietnam (18 (0.2%))

Rule: Emitted as origin:Vietnam when the text has from Vietnam, with a comma after the country, or a demonym for Vietnam followed immediately by a person-origin noun.

```

226: "Vietnam",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!({_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.13. origin:Haiti (16 (0.2%))

Rule: Emitted as origin:Haiti when the text has from Haiti, with a comma after the country, or a demonym for Haiti followed immediately by a person-origin noun.

```

116: "Haiti",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.14. origin:Afghanistan (15 (0.1%))

Rule: Emitted as origin:Afghanistan when the text has from Afghanistan, with a comma after the country, or a demonym for Afghanistan followed immediately by a person-origin noun.

```

50:     "Afghanistan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.15. origin:Dominican-Republic (14 (0.1%))

Rule: Emitted as origin:Dominican-Republic when the text has from Dominican Republic, with a comma after the country, or a demonym for Dominican Republic followed immediately by a person-origin noun.

```

95: "Dominican Republic",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{ _PLACE },\s+({ _PLACE })\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.16. origin:Nicaragua (14 (0.1%))

Rule: Emitted as origin:Nicaragua when the text has from Nicaragua, with a comma after the country, or a demonym for Nicaragua followed immediately by a person-origin noun.

```

167: "Nicaragua",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.17. origin:India (13 (0.1%))

Rule: Emitted as origin:India when the text has from India, with a comma after the country, or a demonym for India followed immediately by a person-origin noun.

```

120:     "India",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.18. origin:Jamaica (13 (0.1%))

Rule: Emitted as origin:Jamaica when the text has from Jamaica, with a comma after the country, or a demonym for Jamaica followed immediately by a person-origin noun.

```

127: "Jamaica",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.19. origin:Somalia (13 (0.1%))

Rule: Emitted as origin:Somalia when the text has from Somalia, with a comma after the country, or a demonym for Somalia followed immediately by a person-origin noun.

```

192: "Somalia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*," )
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.20. origin:Brazil (12 (0.1%))

Rule: Emitted as origin:Brazil when the text has from Brazil, with a comma after the country, or a demonym for Brazil followed immediately by a person-origin noun.

```

72: "Brazil",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.21. origin:Peru (6 (0.1%))

Rule: Emitted as origin:Peru when the text has from Peru, with a comma after the country, or a demonym for Peru followed immediately by a person-origin noun.

```

176: "Peru",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.22. origin:Guyana (5 (0.0%))

Rule: Emitted as origin:Guyana when the text has from Guyana, with a comma after the country, or a demonym for Guyana followed immediately by a person-origin noun.

```

115: "Guyana",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.23. origin:Nigeria (5 (0.0%))

Rule: Emitted as origin:Nigeria when the text has from Nigeria, with a comma after the country, or a demonym for Nigeria followed immediately by a person-origin noun.

```

169: "Nigeria",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.24. origin:Russia (5 (0.0%))

Rule: Emitted as origin:Russia when the text has from Russia, with a comma after the country, or a demonym for Russia followed immediately by a person-origin noun.

```

182: "Russia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.25. origin:Ukraine (5 (0.0%))

Rule: Emitted as origin:Ukraine when the text has from Ukraine, with a comma after the country, or a demonym for Ukraine followed immediately by a person-origin noun.

```

215: "Ukraine",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.26. origin:Cambodia (4 (0.0%))

Rule: Emitted as origin:Cambodia when the text has from Cambodia, with a comma after the country, or a demonym for Cambodia followed immediately by a person-origin noun.

```

77: "Cambodia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.27. origin:Syria (4 (0.0%))

Rule: Emitted as origin:Syria when the text has from Syria, with a comma after the country, or a demonym for Syria followed immediately by a person-origin noun.

```

202:     "Syria",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.28. origin:Kenya (3 (0.0%))

Rule: Emitted as origin:Kenya when the text has from Kenya, with a comma after the country, or a demonym for Kenya followed immediately by a person-origin noun.

```

131: "Kenya",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country: {_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country: {_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin: {_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.29. origin:Sierra-Leone (3 (0.0%))

Rule: Emitted as origin:Sierra-Leone when the text has from Sierra Leone, with a comma after the country, or a demonym for Sierra Leone followed immediately by a person-origin noun.

```

188: "Sierra Leone",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.30. origin:Sudan (3 (0.0%))

Rule: Emitted as origin:Sudan when the text has from Sudan, with a comma after the country, or a demonym for Sudan followed immediately by a person-origin noun.

```

198: "Sudan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.31. origin:Angola (2 (0.0%))

Rule: Emitted as origin:Angola when the text has from Angola, with a comma after the country, or a demonym for Angola followed immediately by a person-origin noun.

```

54: "Angola",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.32. origin:Barbados (2 (0.0%))

Rule: Emitted as origin:Barbados when the text has from Barbados, with a comma after the country, or a demonym for Barbados followed immediately by a person-origin noun.

```

63: "Barbados",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!({_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{[_PREPOSITIONS]})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.33. origin:Canada (2 (0.0%))

Rule: Emitted as origin:Canada when the text has from Canada, with a comma after the country, or a demonym for Canada followed immediately by a person-origin noun.

```

79: "Canada",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.34. origin:Chile (2 (0.0%))

Rule: Emitted as origin:Chile when the text has from Chile, with a comma after the country, or a demonym for Chile followed immediately by a person-origin noun.

```

81:     "Chile",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.35. origin:Egypt (2 (0.0%))

Rule: Emitted as origin:Egypt when the text has from Egypt, with a comma after the country, or a demonym for Egypt followed immediately by a person-origin noun.

```

97: "Egypt",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.36. origin:Ghana (2 (0.0%))

Rule: Emitted as origin:Ghana when the text has from Ghana, with a comma after the country, or a demonym for Ghana followed immediately by a person-origin noun.

```

110:     "Ghana",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.37. origin:Iraq (2 (0.0%))

Rule: Emitted as origin:Iraq when the text has from Iraq, with a comma after the country, or a demonym for Iraq followed immediately by a person-origin noun.

```

123:     "Iraq",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.38. origin:Pakistan (2 (0.0%))

Rule: Emitted as origin:Pakistan when the text has from Pakistan, with a comma after the country, or a demonym for Pakistan followed immediately by a person-origin noun.

```

172: "Pakistan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.39. origin:Panama (2 (0.0%))

Rule: Emitted as origin:Panama when the text has from Panama, with a comma after the country, or a demonym for Panama followed immediately by a person-origin noun.

```

174: "Panama",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.40. origin:Philippines (2 (0.0%))

Rule: Emitted as origin:Philippines when the text has from Philippines, with a comma after the country, or a demonym for Philippines followed immediately by a person-origin noun.

```

177: "Philippines",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.41. origin:Thailand (2 (0.0%))

Rule: Emitted as origin:Thailand when the text has from Thailand, with a comma after the country, or a demonym for Thailand followed immediately by a person-origin noun.

```

206: "Thailand",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE})\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.42. origin:Turkey (2 (0.0%))

Rule: Emitted as origin:Turkey when the text has from Turkey, with a comma after the country, or a demonym for Turkey followed immediately by a person-origin noun.

```

212: "Turkey",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.43. origin:Argentina (1 (0.0%))

Rule: Emitted as origin:Argentina when the text has from Argentina, with a comma after the country, or a demonym for Argentina followed immediately by a person-origin noun.

```

55: "Argentina",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!({_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.44. origin:Azerbaijan (1 (0.0%))

Rule: Emitted as origin:Azerbaijan when the text has from Azerbaijan, with a comma after the country, or a demonym for Azerbaijan followed immediately by a person-origin noun.

```

59: "Azerbaijan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.45. origin:Bangladesh (1 (0.0%))

Rule: Emitted as origin:Bangladesh when the text has from Bangladesh, with a comma after the country, or a demonym for Bangladesh followed immediately by a person-origin noun.

```

62: "Bangladesh",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.46. origin:Burkina-Faso (1 (0.0%))

Rule: Emitted as origin:Burkina-Faso when the text has from Burkina Faso, with a comma after the country, or a demonym for Burkina Faso followed immediately by a person-origin noun.

```

75: "Burkina Faso",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.47. origin:Cameroon (1 (0.0%))

Rule: Emitted as origin:Cameroon when the text has from Cameroon, with a comma after the country, or a demonym for Cameroon followed immediately by a person-origin noun.

```

78: "Cameroon",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.48. origin:Costa-Rica (1 (0.0%))

Rule: Emitted as origin:Costa-Rica when the text has from Costa Rica, with a comma after the country, or a demonym for Costa Rica followed immediately by a person-origin noun.

```

86: "Costa Rica",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.49. origin:Germany (1 (0.0%))

Rule: Emitted as origin:Germany when the text has from Germany, with a comma after the country, or a demonym for Germany followed immediately by a person-origin noun.

```

109: "Germany",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.50. origin:Ireland (1 (0.0%))

Rule: Emitted as origin:Ireland when the text has from Ireland, with a comma after the country, or a demonym for Ireland followed immediately by a person-origin noun.

```

124: "Ireland",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.51. origin:Israel (1 (0.0%))

Rule: Emitted as origin:Israel when the text has from Israel, with a comma after the country, or a demonym for Israel followed immediately by a person-origin noun.

```

125:     "Israel",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.52. origin:Italy (1 (0.0%))

Rule: Emitted as origin:Italy when the text has from Italy, with a comma after the country, or a demonym for Italy followed immediately by a person-origin noun.

```

126:     "Italy",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.53. origin:Kazakhstan (1 (0.0%))

Rule: Emitted as origin:Kazakhstan when the text has from Kazakhstan, with a comma after the country, or a demonym for Kazakhstan followed immediately by a person-origin noun.

```

130:     "Kazakhstan",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.54. origin:Latvia (1 (0.0%))

Rule: Emitted as origin:Latvia when the text has from Latvia, with a comma after the country, or a demonym for Latvia followed immediately by a person-origin noun.

```

137: "Latvia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS_})\b)[A-Za-z][a-zA-Z]*)"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS_})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.55. origin:Liberia (1 (0.0%))

Rule: Emitted as origin:Liberia when the text has from Liberia, with a comma after the country, or a demonym for Liberia followed immediately by a person-origin noun.

```

140:     "Liberia",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE },\s+({ _PLACE })\b)")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.56. origin:Mauritania (1 (0.0%))

Rule: Emitted as origin:Mauritania when the text has from Mauritania, with a comma after the country, or a demonym for Mauritania followed immediately by a person-origin noun.

```

151: "Mauritania",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{ _PREPOSITIONS }\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({ _PLACE })\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{ _PREPOSITIONS })\s+({ _PLACE })\b")
1229: # "<Place>," <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({ _PLACE }),\s+({ _PLACE })\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.57. origin:Myanmar (1 (0.0%))

Rule: Emitted as origin:Myanmar when the text has from Myanmar, with a comma after the country, or a demonym for Myanmar followed immediately by a person-origin noun.

```

160:     "Myanmar",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.58. origin:Rwanda (1 (0.0%))

Rule: Emitted as origin:Rwanda when the text has from Rwanda, with a comma after the country, or a demonym for Rwanda followed immediately by a person-origin noun.

```

183: "Rwanda",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?!(?:{_PREPOSITIONS})\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.59. origin:Senegal (1 (0.0%))

Rule: Emitted as origin:Senegal when the text has from Senegal, with a comma after the country, or a demonym for Senegal followed immediately by a person-origin noun.

```

185: "Senegal",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). ` _resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{?i:{_PREPOSITIONS}}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{?i:{_PREPOSITIONS}})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```



```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

9.60. origin:United-Kingdom (1 (0.0%))

Rule: Emitted as origin:United-Kingdom when the text has from United Kingdom, with a comma after the country, or a demonym for United Kingdom followed immediately by a person-origin noun.

```

218: "United Kingdom",

1217: # second word excludes those same prepositions via lookahead, so a greedy
1218: # capture can't swallow the preposition that introduces the *next* place
1219: # ("from USA to Mexico" -> "USA" + "Mexico", not "USA to"). `_resolve_vocab`
1220: # then validates against the vocab and falls back to the leading word for any
1221: # other trailing token. Kept conservative because raw country names
1222: # false-positive easily ("Chad", "Georgia").
1223: _PREPOSITIONS = "from|in|to|of|with|by"
1224: _PLACE = rf"[A-Za-z][a-zA-Z]*(?:\s+(?:{:_PREPOSITIONS}\b)[A-Za-z][a-zA-Z]+)?"
1225: # "from <Country>," – anchors the ORIGIN validator.
1226: PATTERN_ORIGIN_CANDIDATE = re.compile(rf"\b(?:from)\s+({_PLACE})\s*,")
1227: # Any sovereign-state mention after a preposition.
1228: PATTERN_COUNTRY_CANDIDATE = re.compile(rf"\b(?:{:_PREPOSITIONS})\s+({_PLACE})\b")
1229: # "<Place>,<State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b({_PLACE}),\s+({_PLACE})\b")

1238: DEMONYM_TO_COUNTRY: dict[str, str] = {
1239:     "iranian": "Iran",
1240:     "mexican": "Mexico",
1241:     "venezuelan": "Venezuela",
1242:     "chinese": "China",
1243:     "russian": "Russia",
1244:     "cuban": "Cuba",
1245:     "colombian": "Colombia",
1246:     "salvadoran": "El Salvador",
1247:     "salvadorian": "El Salvador",
1248:     "honduran": "Honduras",
1249:     "guatemalan": "Guatemala",
1250:     "haitian": "Haiti",
1251:     "somali": "Somalia",
1252:     "somalian": "Somalia",
1253:     "afghan": "Afghanistan",
1254:     "syrian": "Syria",

```

```

1255: "nigerian": "Nigeria",
1256: "pakistani": "Pakistan",
1257: "ukrainian": "Ukraine",
1258: "israeli": "Israel",
1259: "iraqi": "Iraq",
1260: "egyptian": "Egypt",
1261: "yemeni": "Yemen",
1262: "brazilian": "Brazil",
1263: "nicaraguan": "Nicaragua",
1264: "ecuadorian": "Ecuador",
1265: "peruvian": "Peru",
1266: "jamaican": "Jamaica",
1267: "filipino": "Philippines",
1268: "vietnamese": "Vietnam",
1269: "lebanese": "Lebanon",
1270: "libyan": "Libya",
1271: "sudanese": "Sudan",
1272: "ethiopian": "Ethiopia",
1273: "kenyan": "Kenya",
1274: "moroccan": "Morocco",
1275: "algerian": "Algeria",
1276: "bangladeshi": "Bangladesh",
1277: "cambodian": "Cambodia",
1278: "indonesian": "Indonesia",
1279: "dominican": "Dominican Republic",
1280: "panamanian": "Panama",
1281: "bolivian": "Bolivia",
1282: "chilean": "Chile",
1283: "argentine": "Argentina",
1284: "argentinian": "Argentina",
1285: "saudi": "Saudi Arabia",
1286: "turkish": "Turkey",
1287: }
1288: FOREIGN_CITY_TO_COUNTRY: dict[str, str] = {

```

```

1289: "tehran": "Iran",
1290: "caracas": "Venezuela",
1291: "beijing": "China",
1292: "moscow": "Russia",
1293: "havana": "Cuba",
1294: "kabul": "Afghanistan",
1295: "damascus": "Syria",
1296: "baghdad": "Iraq",
1297: "tripoli": "Libya",
1298: "mogadishu": "Somalia",
1299: "bogota": "Colombia",
1300: "managua": "Nicaragua",
1301: "tegucigalpa": "Honduras",
1302: "kyiv": "Ukraine",
1303: }
1304:
1305:
1306: def _alternation_pattern(keys: list[str]) -> re.Pattern[str]:
1307:     alts = sorted((re.escape(k) for k in keys), key=len, reverse=True)
1308:     return re.compile(r"\b(" + "|".join(alts) + r")\b", re.IGNORECASE)
1309:
1310:
1311: PATTERN_DEMONYM = _alternation_pattern(list(DEMONYM_TO_COUNTRY))
1312: PATTERN_FOREIGN_CITY = _alternation_pattern(list(FOREIGN_CITY_TO_COUNTRY))
1313: # A demonym that immediately precedes one of these nouns describes a person's
1314: # nationality, so it also earns origin:<country>, not just a contextual country:.
1315: PATTERN_DEMONYM_PERSON = _compile(
1316:     r"^\W*(?:national|nationals|citizen|citizens|immigrant|immigrants|migrant|migrants|"

1677: # origin:<COUNTRY> – validated against the sovereign-state vocab.
1678: for m in PATTERN_ORIGIN_CANDIDATE.finditer(text):
1679:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1680:     if resolved:
1681:         add(f"origin:{_normalize_country(resolved)}", span=m.span(1))

```

```

1682:
1683: # country:<NAME> – broader: any contextual country mention.
1684: for m in PATTERN_COUNTRY_CANDIDATE.finditer(text):
1685:     resolved = _resolve_vocab(m.group(1) or "", COUNTRY_BY_LOWER)
1686:     if resolved:
1687:         add(f"country:{_normalize_country(resolved)}", span=m.span(1))
1688:
1689: # Demonyms ("Iranian regime") and major foreign cities ("in Tehran") ->
1690: # country, which the bare-name validators above can't see. A demonym
1691: # followed by a person noun also earns origin:.
1692: for m in PATTERN_DEMONYM.finditer(text):
1693:     country = DEMONYM_TO_COUNTRY.get(m.group(1).lower())
1694:     if country and country.lower() in COUNTRY_LOWER:
1695:         add(f"country:{_normalize_country(country)}", span=m.span(1))
1696:         if PATTERN_DEMONYM_PERSON.search(text[m.end() : m.end() + 24]):
1697:             add(f"origin:{_normalize_country(country)}", span=m.span(1))
1698: for m in PATTERN_FOREIGN_CITY.finditer(text):
1699:
2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
2052: if low == "burma":
2053:     return "Myanmar"
2054: return s.replace(" ", "-")
2055:
2056:
2057: def _normalize_state(s: str) -> str:
2058:     return s.replace(" ", "-")
2059:
2060:
2061: def _resolve_vocab(candidate: str, vocab: dict[str, str]) -> str | None:

```

```

2062: """Resolve a candidate (any case, possibly two words) to its canonical
2063: vocab spelling.
2064:
2065: Tries the full phrase first, then its leading word, so a greedy
2066: case-insensitive capture that swallowed a trailing lowercase token

```

10. parody

The parody namespace is produced by the franchise gazetteer. A franchise match adds both `genre:parody` and `parody:<franchise>`. Star Wars also has a two-cue rule: this is the way plus at least one other Star Wars cue.

10.1. parody:star-wars (2 (0.0%))

Rule: Matches May the 4th/fourth be with you, force-with-you language, Jedi/Sith/Death Star/lightsaber/Darth Vader/Skywalker, or galaxy/republic/empire/rebellion/Mandalorian within 200 characters of Jedi/Sith/Death Star/the way. Also matches this is the way plus another Star Wars cue.

```

850: PARODY_FRANCHISE_GAZETTEER: tuple[tuple[re.Pattern[str], str], ...] = (
851:     (
852:         _compile(
853:             r"\bmay\s+the\s+(?:4th|fourth)\s+be\s+with\s+you\b"
854:             r"|\bthe\s+force\s+(?:is|be|will\s+be)\s+with\b"
855:             r"|\b(?:jedi\sith|death\s+star|lightsaber|darth\s+vader|skywalker)\b"
856:             r"|\b(?:galaxy|republic|empire|rebellion|mandalorian)\b.{0,200}\b(?:jedi\sith|death\s+star|the\s+way)\b"
857:         ),
858:         "star-wars",
859:     ),
860:     (
861:         _compile(
1656:             if m := franchise_pat.search(text):
1657:                 add("genre:parody", span=m.span())
1658:                 add(f"parody:{franchise_slug}", span=m.span())
1659:                 add("theme:pop-culture-reference")
1660:             # Also catch "this is the way" + one more Star Wars cue (e.g. "a galaxy").
1661:             if (
1662:                 PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1663:                 and PATTERN_PARODY_STAR_WARS_EXTRA_CUE.search(text)
1664:                 and not any(e["tag"] == "parody:star-wars" for e in entries)
1665:             ):
1666:                 m_way = PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1667:                 if m_way:
1668:                     add("genre:parody", span=m_way.span())
1669:                     add("parody:star-wars", span=m_way.span())
1670:                     add("theme:pop-culture-reference")

```

10.2. parody:game-of-thrones (1 (0.0%))

Rule: Matches winter is coming, Game of Thrones, the iron throne, or you know nothing Jon Snow.

```

871:         ),
872:         "top-gun",
873:     ),
874:     (
875:         _compile(
876:             r"\bwinter\s+is\s+coming\b|\bgame\s+of\s+thrones\b|\bthe\s+iron\s+throne\b"
877:             r"|\byou\s+know\s+nothing,?\s+jon\s+snow\b"
878:         ),
879:         "game-of-thrones",
880:     ),
881:     (
882:         _compile(
1656:             if m := franchise_pat.search(text):
1657:                 add("genre:parody", span=m.span())
1658:                 add(f"parody:{franchise_slug}", span=m.span())
1659:                 add("theme:pop-culture-reference")
1660:             # Also catch "this is the way" + one more Star Wars cue (e.g. "a galaxy").
1661:             if (
1662:                 PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1663:                 and PATTERN_PARODY_STAR_WARS_EXTRA_CUE.search(text)
1664:                 and not any(e["tag"] == "parody:star-wars" for e in entries)
1665:             ):
1666:                 m_way = PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1667:                 if m_way:
1668:                     add("genre:parody", span=m_way.span())
1669:                     add("parody:star-wars", span=m_way.span())
1670:                     add("theme:pop-culture-reference")

```

10.3. parody:ghostbusters (1 (0.0%))

Rule: Matches who you/ya gonna call or Ghostbusters.

```

906:         _compile(
907:             r"\bavengers\s+assemble\b|\bwakanda\s+forever\b|\binfinity\s+(?:gauntlet|stones?)\b"
908:             r"|\bi\s+am\s+iron\s+man\b|\bthanos\b"
909:         ),
910:         "marvel",
911:     ),
912:     (
913:         _compile(r"\bwho\s+(?:you|ya)\s+gonna\s+call\b|\bghostbusters\b"),
914:         "ghostbusters",
915:     ),
916:     (
917:         _compile(r"\boffer\s+(?:he|you|they)\s+can'?t\s+refuse\b|\bthe\s+godfather\b"),
1656:             if m := franchise_pat.search(text):
1657:                 add("genre:parody", span=m.span())
1658:                 add(f"parody:{franchise_slug}", span=m.span())
1659:                 add("theme:pop-culture-reference")
1660:             # Also catch "this is the way" + one more Star Wars cue (e.g. "a galaxy").
1661:             if (
1662:                 PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1663:                 and PATTERN_PARODY_STAR_WARS_EXTRA_CUE.search(text)
1664:                 and not any(e["tag"] == "parody:star-wars" for e in entries)
1665:             ):
1666:                 m_way = PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1667:                 if m_way:
1668:                     add("genre:parody", span=m_way.span())
1669:                     add("parody:star-wars", span=m_way.span())
1670:                     add("theme:pop-culture-reference")

```

10.4. parody:pawn-stars (1 (0.0%))

Rule: Matches Pawn Stars.

```

914:         "ghostbusters",
915:     ),
916:     (
917:         _compile(r"\boffer\s+(?:he|you|they)\s+can'?t\s+refuse\b|\bthe\s+godfather\b"),
918:         "godfather",
919:     ),
920:     (
921:         _compile(r"\bpawn\s+stars\b"),
922:         "pawn-stars",
923:     ),
924: )
925: # Emit genre:parody ONLY when a franchise match fires; it never fires alone.
1656:             if m := franchise_pat.search(text):
1657:                 add("genre:parody", span=m.span())
1658:                 add(f"parody:{franchise_slug}", span=m.span())
1659:                 add("theme:pop-culture-reference")
1660:             # Also catch "this is the way" + one more Star Wars cue (e.g. "a galaxy").
1661:             if (
1662:                 PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1663:                 and PATTERN_PARODY_STAR_WARS_EXTRA_CUE.search(text)
1664:                 and not any(e["tag"] == "parody:star-wars" for e in entries)
1665:             ):
1666:                 m_way = PATTERN_PARODY_STAR_WARS_MULTI.search(text)
1667:                 if m_way:
1668:                     add("genre:parody", span=m_way.span())
1669:                     add("parody:star-wars", span=m_way.span())
1670:                     add("theme:pop-culture-reference")

```

11. phrase

The phrase namespace is produced by literal regexes for migrant(s) and immigrant(s). Each phrase also implies topic:immigration.

11.1. phrase:immigrant (234 (2.3%))

Rule: Matches immigrant/immigrants.

```
1177: PATTERN_PHRASE_IMMIGRANT = _compile(r"\bimmigrants?\b")
1587: (PATTERN_PHRASE_IMMIGRANT, "phrase:immigrant"),
```

11.2. phrase:migrant (142 (1.4%))

Rule: Matches migrant/migrants.

```
1176: PATTERN_PHRASE_MIGRANT = _compile(r"\bmigrants?\b")
1586: (PATTERN_PHRASE_MIGRANT, "phrase:migrant"),
```

12. policy

The policy namespace is produced by regexes for named policy areas: border, sanctuary, worksite enforcement, and CBP Home/self-deportation.

12.1. policy:border (1,071 (10.7%))

Rule: Matches border, southwest border, crossing, or border wall.

```
652: PATTERN_THEME_BORDER = _compile(r"\b(border|southwest border|crossing|border wall)\b")
1552: (PATTERN_THEME_BORDER, "policy:border"),
```

12.2. policy:cbp-home (236 (2.4%))

Rule: Matches CBP Home, Project Homecoming, self-deport variants, free/complimentary ticket/flight home language, exit bonus, take control of departure, or go home within 120 characters of alien/migrant/CBP Home/self-deport/deport context.

```
786: PATTERN_THEME_CBP_HOME = _compile(
787:     r"\bCBP\s+Home\b"
788:     r"|\bProject\s+Homecoming\b"
789:     r"|\bself[- ]deport(?:ed|ing|ation)?\b"
790:     r"|\bfree\s+(?:flight|plane\s+ticket|ticket)\s+home\b"
791:     r"|\bcomplimentary\s+plane\s+ticket\s+home\b"
792:     r"|\bexit\s+bonus\b"
793:     r"|\btake\s+control\s+of\s+(?:your|their|his|her)\s+departure\b"
794:     r"|\b(?:illegal\s+aliens?|aliens?|migrants?|CBP\s+Home|self[- ]deport|deport)\b"
795:     r"[\s]{0,120}\bgo\s+home\b"
796:     r"|\bgo\s+home\b[\s]{0,120}\b"
797:     r"?(?:illegal\s+aliens?|aliens?|migrants?|CBP\s+Home|self[- ]deport|deport)\b"
798: )
1560: (PATTERN_THEME_CBP_HOME, "policy:cbp-home"),
```

12.3. policy:sanctuary-cities (191 (1.9%))

Rule: Matches sanctuary city/cities, sanctuary jurisdiction, sanctuary policy/policies, or sanctuary state.

```
653: PATTERN_THEME_SANCTUARY = _compile(
654:     r"\b(sanctuary (?:cit(?:y|ies)|jurisdiction|polic(?:y|ies))|sanctuary state)\b"
655: )
1553: (PATTERN_THEME_SANCTUARY, "policy:sanctuary-cities"),
```

12.4. policy:worksite-enforcement (21 (0.2%))

Rule: Matches worksite, workplace, I-9, E-Verify, employer audit, or employer raid.

```

656: PATTERN_THEME_WORKSITE = _compile(r"\b(worksite|workplace|I-9|E-Verify|employer (?:(audit|raid))\b)")
1554:         (PATTERN_THEME_WORKSITE, "policy:worksite-enforcement"),

```

13. religion

The religion namespace in this workbook contains Christianity. It is produced by a Christian-specific regex over the combined text string.

13.1. religion:christianity (85 (0.8%))

Rule: Matches Christian/Christianity/Christians, Judeo-Christian, Christian faith/values/church/heritage/nation, Jesus/Jesus Christ, Christ is king/our lord, Bible/biblical/scripture/scriptural, Bible book chapter:verse references, or in Jesus' name.

```

681: PATTERN_THEME_CHRISTIANITY = _compile(
682:     r"\bchristian(?:ity|s)?\b"
683:     r"|\bjudeo[- ]christian\b"
684:     r"|\bchristian\s+(?:faith|values?|church|churches|heritage|nation)\b"
685:     r"|\bjesus(?:\s+christ)?\b"
686:     r"|\bchrist\s+(?:is\s+king|the\s+king|our\s+lord)\b"
687:     r"|\b(?:biblical|scripture|scriptural)\b"
688:     r"|\b(?:[1-3]\s+)?(?:Genesis|Exodus|Leviticus|Numbers|Deuteronomy|Joshua|Judges|Ruth|"
689:     r"Samuel|Kings|Chronicles|Ezra|Nehemiah|Esther|Job|Psalms|Proverbs|Ecclesiastes|"
690:     r"Song\s+of\s+Songs|Isaiah|Jeremiah|Lamentations|Ezekiel|Daniel|Hosea|Joel|Amos|"
691:     r"Obadiah|Jonah|Micah|Nahum|Habakkuk|Zephaniah|Haggai|Zechariah|Malachi|Matthew|"
692:     r"Mark|Luke|John|Acts|Romans|Corinthians|Galatians|Ephesians|Philippians|"
693:     r"Colossians|Thessalonians|Timothy|Titus|Philemon|Hebrews|James|Peter|Jude|"
694:     r"Revelation)\s+\d{1,3}:\d{1,3}(?:[-\u2013]\d{1,3})?\b"
695:     r"|\bin\s+jesus(?:'s')?\s+name\b"
696: )
1557:         (PATTERN_THEME_CHRISTIANITY, "religion:christianity"),

```

14. slogan

The slogan namespace is produced by individual recurring-phrase regexes. Some are literal phrases; some require nearby context, such as go home within 120 characters of immigration/deportation language.

14.1. slogan:illegal-alien (2,287 (22.9%))

Rule: Matches illegal alien or illegal aliens.

```

1149: PATTERN_SLOGAN_ILLEGAL_ALIEN = _compile(r"\billegal\s+aliens?\b")
1568:         (PATTERN_SLOGAN_ILLEGAL_ALIEN, "slogan:illegal-alien"),

```

14.2. slogan:criminal-illegal-alien (1,402 (14.0%))

Rule: Matches criminal illegal alien or criminal illegal aliens.

```

1148: PATTERN_SLOGAN_CRIMINAL_ILLEGAL_ALIEN = _compile(r"\bcriminal\s+illegal\s+aliens?\b")
1567:         (PATTERN_SLOGAN_CRIMINAL_ILLEGAL_ALIEN, "slogan:criminal-illegal-alien"),

```

14.3. slogan:worst (222 (2.2%))

Rule: Matches the literal phrase WORST OF THE WORST.

```

1146: PATTERN_SLOGAN_WORST = _compile(r"\bWORST OF THE WORST\b")
1565:         (PATTERN_SLOGAN_WORST, "slogan:worst"),

```

14.4. slogan:masa (140 (1.4%))

Rule: Matches Make America Safe Again.

```

1164: PATTERN_SLOGAN_MASA = _compile(r"\bMake\s+America\s+Safe\s+Again\b")
1574:         (PATTERN_SLOGAN_MASA, "slogan:masa"),

```

14.5. slogan:america-first (72 (0.7%))

Rule: Matches America First or AmericaFirst.

```
1165: PATTERN_SLOGAN_AMERICA_FIRST = _compile(r"\bAmerica\s+First\b|\bAmericaFirst\b")
1575: (PATTERN_SLOGAN_AMERICA_FIRST, "slogan:america-first"),
```

14.6. slogan:maga (58 (0.6%))

Rule: Matches MAGA or Make America Great Again.

```
1162: PATTERN_SLOGAN_MAGA = _compile(r"\bMAGA\b|\bMake\s+America\s+Great\s+Again\b")
1572: (PATTERN_SLOGAN_MAGA, "slogan:maga"),
```

14.7. slogan:law-and-order (51 (0.5%))

Rule: Matches Law and Order or Law & Order.

```
1168: PATTERN_SLOGAN_LAW_AND_ORDER = _compile(r"\bLaw\s+and\s+Order\b|\bLaw\s*&\s*Order\b")
1578: (PATTERN_SLOGAN_LAW_AND_ORDER, "slogan:law-and-order"),
```

14.8. slogan:promises-kept (50 (0.5%))

Rule: Matches Promises Made, Promises Kept, with comma/semicolon/colon flexibility, or Promises Kept.

```
1170: PATTERN_SLOGAN_PROMISES_KEPT = _compile(
1171:     r"\bPromises?\s+Made[,;:]?\s+Promises?\s+Kept\b|\bPromises?\s+Kept\b"
1172: )
1580: (PATTERN_SLOGAN_PROMISES_KEPT, "slogan:promises-kept"),
```

14.9. slogan:most-secure-border (45 (0.4%))

Rule: Matches most secure border or most secured border.

```
1174: PATTERN_SLOGAN_MOST_SECURE_BORDER = _compile(r"\bmost\s+secure(?:d)?\s+border\b")
1582: (PATTERN_SLOGAN_MOST_SECURE_BORDER, "slogan:most-secure-border"),
```

14.10. slogan:golden-age (42 (0.4%))

Rule: Matches Golden Age or Welcome to the Golden Age.

```
1166: PATTERN_SLOGAN_GOLDEN_AGE = _compile(r"\bGolden\s+Age\b|\bWelcome\s+to\s+the\s+Golden\s+Age\b")
1576: (PATTERN_SLOGAN_GOLDEN_AGE, "slogan:golden-age"),
```

14.11. slogan:free-ticket-home (26 (0.3%))

Rule: Matches free ticket home, free flight home, free plane ticket home, complimentary plane ticket home, or free flight to your/their/his/her home country.

```
1150: PATTERN_SLOGAN_FREE_TICKET_HOME = _compile(
1151:     r"\bfree\s+(?:ticket|flight|plane\s+ticket)\s+home\b"
1152:     r"|\bcomplimentary\s+plane\s+ticket\s+home\b"
1153:     r"|\bfree\s+flight\s+to\s+(?:your|their|his|her)\s+home\s+country\b"
1154: )
1569: (PATTERN_SLOGAN_FREE_TICKET_HOME, "slogan:free-ticket-home"),
```

14.12. slogan:mass-deportation (26 (0.3%))

Rule: Matches Mass Deportation or Mass Deportations.

```
1173: PATTERN_SLOGAN_MASS_DEPORTATION = _compile(r"\bMass\s+Deportations?\b")
1581: (PATTERN_SLOGAN_MASS_DEPORTATION, "slogan:mass-deportation"),
```

14.13. slogan:save-america (24 (0.2%))

Rule: Matches Save America or SAVEAMERICA.

```
1167: PATTERN_SLOGAN_SAVE_AMERICA = _compile(r"\bSave\s+America\b|\bSAVEAMERICA\b")
1577: (PATTERN_SLOGAN_SAVE_AMERICA, "slogan:save-america"),
```

14.14. slogan:go-home (16 (0.2%))

Rule: Matches go home only when it appears within 120 characters before or after immigration/deportation context: illegal alien(s), alien(s), migrant(s), CBP Home, self-deport, or deport.

```
1155: PATTERN_SLOGAN_GO_HOME = _compile(
1156:     r"\b(?:illegal\s+aliens?|aliens?|migrants?|CBP\s+Home|self[- ]deport|deport)\b"
1157:     r"[\s\S]{0,120}\bgo\s+home\b"
1158:     r"|\bgo\s+home\b[\s\S]{0,120}\b"
1159:     r"(?:illegal\s+aliens?|aliens?|migrants?|CBP\s+Home|self[- ]deport|deport)\b"
1160: )
1570: (PATTERN_SLOGAN_GO_HOME, "slogan:go-home"),
```

14.15. slogan:peace-through-strength (12 (0.1%))

Rule: Matches Peace Through Strength.

```
1169: PATTERN_SLOGAN_PEACE_THROUGH_STRENGTH = _compile(r"\bPeace\s+Through\s+Strength\b")
1579: (PATTERN_SLOGAN_PEACE_THROUGH_STRENGTH, "slogan:peace-through-strength"),
```

14.16. slogan:catch-release (11 (0.1%))

Rule: Matches catch-and-release or catch and release.

```
1175: PATTERN_SLOGAN_CATCH_RELEASE = _compile(r"\bcatch(?:-and-|\s+and\s+)+release\b")
1583: (PATTERN_SLOGAN_CATCH_RELEASE, "slogan:catch-release"),
```

14.17. slogan:import-third-world (11 (0.1%))

Rule: Matches import the Third World or if you import the Third World.

```
838: PATTERN_SLOGAN_IMPORT_THIRD_WORLD = _compile(
839:     r"\bimport\s+the\s+third\s+world\b"
840:     r"|\bif\s+you\s+import\s+the\s+third\s+world\b"
841: )
1585: (PATTERN_SLOGAN_IMPORT_THIRD_WORLD, "slogan:import-third-world"),
```

14.18. slogan:maha (11 (0.1%))

Rule: Matches MAHA or Make America Healthy Again.

```
1163: PATTERN_SLOGAN_MAHA = _compile(r"\bMAHA\b|\bMake\s+America\s+Healthy\s+Again\b")
1573: (PATTERN_SLOGAN_MAHA, "slogan:maha"),
```

14.19. slogan:nice (9 (0.1%))

Rule: Matches NICE day, NICE morning, ICE is NICE, or NICE city.

```
1145: PATTERN_SLOGAN_NICE = _compile(r"\b(NICE day|NICE morning|ICE is NICE|NICE city)\b")
1564: (PATTERN_SLOGAN_NICE, "slogan:nice"),
```

14.20. slogan:project-homecoming (4 (0.0%))

Rule: Matches Project Homecoming.


```

1161: PATTERN_SLOGAN_PROJECT_HOMECOMING = _compile(r"\bProject\s+Homecoming\b")
1571: (PATTERN_SLOGAN_PROJECT_HOMECOMING, "slogan:project-homecoming"),

```

14.21. slogan:find-and-kill (2 (0.0%))

Rule: Matches we will find you and/& kill you, with optional repeated we will before kill, or find you and/& kill you.

```

832: PATTERN_SLOGAN_FIND_AND_KILL = _compile(
833:     r"\bwe\s+will\s+find\s+you\s+(?:and|&)\s+(?:we\s+will\s+)?kill\s+you\b"
834:     r"|\bfind\s+you\s+(?:and|&)\s+kill\s+you\b"
835: )
1584: (PATTERN_SLOGAN_FIND_AND_KILL, "slogan:find-and-kill"),

```

14.22. slogan:reportrecon (2 (0.0%))

Rule: Matches Report. Recon. Raid. with flexible whitespace after the periods.

```

1147: PATTERN_SLOGAN_REPORTRECON = _compile(r"\bReport\.\s*Recon\.\s*Raid\.")
1566: (PATTERN_SLOGAN_REPORTRECON, "slogan:reportrecon"),

```

15. speaker

The speaker namespace is produced only when a known speaker alias appears near speech context. One pattern is alias followed within 100 characters by a speech/interview/announcement/remarks word. The other is a speech/interview/announcement/remarks word followed within 80 characters by the alias.

15.1. speaker:President Trump (392 (3.9%))

Rule: Matches aliases for President Trump only when the alias is within 100 characters before, or 80 characters after, speech/interview/remarks/announcement/quote/briefing context.

```

1122: (
1123:     "President Trump",
1124:     r"President\s+(?:Donald\s+J\.\s+)?Trump|Donald\s+Trump|@?POTUS|@?realDonaldTrump",
1125: ),
1126: (
1127:     "Vice President Vance",
1128:     r"Vice President\s+Vance",
1129: ),
1130: )
1131:
1132: """Return speaker tags only when a named official is tied to speech."""
1133: out: list[tuple[str, tuple[int, int]]] = []
1134: for canonical, alias in SPEAKER_ALIASES:
1135:     patterns = (
1136:         rf"(?!<!\w)({alias})(?!<!\w)\.{{0,100}}\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
1137:         rf"\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b({alias})\b",
1138:         rf".{{0,80}}\b(?:by|from|with|of|featuring|:)?\s*({alias})(?!<!\w)",
1139:     )
1140:     for pattern in patterns:
1141:         match = re.search(pattern, text, re.I)
1142:         if match:
1143:             out.append((f"speaker:{canonical}", match.span(1)))
1144:             break
1145: return out
1146:
1147:
1148:
1149:
1150:
1151:
1152:
1153:
1154:
1155:
1156:
1157:
1158:
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```

15.2. speaker:Vice President Vance (31 (0.3%))

Rule: Matches aliases for Vice President Vance only when the alias is within 100 characters before, or 80 characters after, speech/interview/remarks/announcement/quote/briefing context.

```

1126: (
1127:     "Vice President Vance",
1128:     r"Vice\s+President\s+(?:JD\s+|J\.D\.\s+)?Vance|VP\s+Vance|J\.\s+Vance|@?VP",
1129: ),
1130: (
1131:     "Tom Homan",

2202: """Return speaker tags only when a named official is tied to speech."""
2203: out: list[tuple[str, tuple[int, int]]] = []
2204: for canonical, alias in SPEAKER_ALIASES:
2205:     patterns = (
2206:         rf"(?!w)({alias})(?!w).{{0,100}}\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2207:         rf"\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2208:         rf".{{0,80}}\b(?:by|from|with|of|featuring|:)?s*({alias})(?!w)",
2209:     )
2210:     for pattern in patterns:
2211:         match = re.search(pattern, text, re.I)
2212:         if match:
2213:             out.append((f"speaker:{canonical}", match.span(1)))
2214:             break
2215: return out
2216:
2217:

```

15.3. speaker:Secretary Mullin (22 (0.2%))

Rule: Matches aliases for Secretary Mullin only when the alias is within 100 characters before, or 80 characters after, speech/interview/remarks/announcement/quote/briefing context.

```

1118: (
1119:     "Secretary Mullin",
1120:     r"Secretary\s+Mullin|Sec\.\s+Mullin|@?SecMullinDHS",
1121: ),
1122: (
1123:     "President Trump",

2202: """Return speaker tags only when a named official is tied to speech."""
2203: out: list[tuple[str, tuple[int, int]]] = []
2204: for canonical, alias in SPEAKER_ALIASES:
2205:     patterns = (
2206:         rf"(?!w)({alias})(?!w).{{0,100}}\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2207:         rf"\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2208:         rf".{{0,80}}\b(?:by|from|with|of|featuring|:)?s*({alias})(?!w)",
2209:     )
2210:     for pattern in patterns:
2211:         match = re.search(pattern, text, re.I)
2212:         if match:
2213:             out.append((f"speaker:{canonical}", match.span(1)))
2214:             break
2215: return out
2216:
2217:

```

15.4. speaker:Tom Homan (11 (0.1%))

Rule: Matches aliases for Tom Homan only when the alias is within 100 characters before, or 80 characters after, speech/interview/remarks/announcement/quote/briefing context.

```

1130: (
1131:     "Tom Homan",
1132:     r"Tom\s+Homan|Thomas\s+D\.\s+Homan|@?RealTomHoman",
1133: ),
1134: (
1135:     "Stephen Miller",

2202: """Return speaker tags only when a named official is tied to speech."""
2203: out: list[tuple[str, tuple[int, int]]] = []
2204: for canonical, alias in SPEAKER_ALIASES:
2205:     patterns = (
2206:         rf"(?!w)({alias})(?!w).{{0,100}}\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2207:         rf"\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2208:         rf".{{0,80}}\b(?:by|from|with|of|featuring|:)?s*({alias})(?!w)",
2209:     )
2210:     for pattern in patterns:
2211:         match = re.search(pattern, text, re.I)
2212:         if match:
2213:             out.append((f"speaker:{canonical}", match.span(1)))
2214:             break
2215: return out
2216:
2217:

```

15.5. speaker:First Lady Melania Trump (6 (0.1%))

Rule: Matches aliases for First Lady Melania Trump only when the alias is within 100 characters before, or 80 characters after, speech/interview/remarks/announcement/quote/briefing context.

```

1114: (
1115:     "First Lady Melania Trump",
1116:     r"(?:First\s+Lady\s+)?Melania\s+Trump|FLOTUS",
1117: ),
1118: (
1119:     "Secretary Mullin",

2202: """Return speaker tags only when a named official is tied to speech."""
2203: out: list[tuple[str, tuple[int, int]]] = []
2204: for canonical, alias in SPEAKER_ALIASES:
2205:     patterns = (
2206:         rf"(?!w)({alias})(?!w).{{0,100}}\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2207:         rf"\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2208:         rf".{{0,80}}\b(?:by|from|with|of|featuring|:)?s*({alias})(?!w)",
2209:     )
2210:     for pattern in patterns:
2211:         match = re.search(pattern, text, re.I)
2212:         if match:
2213:             out.append((f"speaker:{canonical}", match.span(1)))
2214:             break
2215: return out
2216:
2217:

```

15.6. speaker:Stephen Miller (3 (0.0%))

Rule: Matches aliases for Stephen Miller only when the alias is within 100 characters before, or 80 characters after, speech/interview/remarks/announcement/quote/briefing context.

```

1134: (
1135:     "Stephen Miller",
1136:     r"Stephen\s+Miller|@?StephenM",
1137: ),
1138: (
1139:     "Gregory Bovino",

2202: """Return speaker tags only when a named official is tied to speech."""
2203: out: list[tuple[str, tuple[int, int]]] = []
2204: for canonical, alias in SPEAKER_ALIASES:
2205:     patterns = (
2206:         rf"(?!w)({alias})(?!w).{{0,100}}\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2207:         rf"\b(?:{SPEAKER_ACTION_CONTEXT}|{SPEAKER_NOUN_CONTEXT})\b",
2208:         rf".{{0,80}}\b(?:by|from|with|of|featuring|:)?s*({alias})(?!w)",
2209:     )
2210:     for pattern in patterns:
2211:         match = re.search(pattern, text, re.I)
2212:         if match:
2213:             out.append((f"speaker:{canonical}", match.span(1)))
2214:             break
2215: return out
2216:
2217:

```

16. state

The state namespace is produced by a validated '<place>', <state>' extraction. The captured state must resolve against the U.S. state vocabulary.

16.1. state:Texas (133 (1.3%))

Rule: Emitted as state:Texas when '<place>', Texas' is captured and the state resolves against the U.S. state vocabulary.

```

277:     "Texas",

1229: # "<Place>, <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> - the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.2. state:California (89 (0.9%))

Rule: Emitted as state:California when '<place>', California' is captured and the state resolves against the U.S. state vocabulary.

```

239:      "California",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.3. state:New-York (57 (0.6%))

Rule: Emitted as state:New-York when '<place>, New York' is captured and the state resolves against the U.S. state vocabulary.

```

266:      "New York",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.4. state:Florida (56 (0.6%))

Rule: Emitted as state:Florida when '<place>, Florida' is captured and the state resolves against the U.S. state vocabulary.

```

243:      "Florida",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.5. state:Virginia (46 (0.5%))

Rule: Emitted as state:Virginia when '<place>, Virginia' is captured and the state resolves against the U.S. state vocabulary.

```

280:      "Virginia",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.6. state:North-Carolina (23 (0.2%))

Rule: Emitted as state:North-Carolina when '<place>, North Carolina' is captured and the state resolves against the U.S. state vocabulary.

```
267:      "North Carolina",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:      # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:      for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:          resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:          if resolved:
1707:              add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
```

16.7. state:Utah (22 (0.2%))

Rule: Emitted as state:Utah when '<place>, Utah' is captured and the state resolves against the U.S. state vocabulary.

```
278:      "Utah",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:      # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:      for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:          resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:          if resolved:
1707:              add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
```

16.8. state:Illinois (21 (0.2%))

Rule: Emitted as state:Illinois when '<place>, Illinois' is captured and the state resolves against the U.S. state vocabulary.

```
247:      "Illinois",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:      # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:      for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:          resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:          if resolved:
1707:              add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:      if low in ("usa", "united states"):
2047:          return "United-States"
2048:      if low == "uk":
2049:          return "United-Kingdom"
2050:      if low == "uae":
2051:          return "United-Arab-Emirates"
```

16.9. state:Tennessee (16 (0.2%))

Rule: Emitted as state:Tennessee when '<place>, Tennessee' is captured and the state resolves against the U.S. state vocabulary.

```

276:      "Tennessee",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"

```

16.10. state:Minnesota (15 (0.1%))

Rule: Emitted as state:Minnesota when '<place>, Minnesota' is captured and the state resolves against the U.S. state vocabulary.

```

257:      "Minnesota",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"

```

16.11. state:Washington (15 (0.1%))

Rule: Emitted as state:Washington when '<place>, Washington' is captured and the state resolves against the U.S. state vocabulary.

```

281:      "Washington",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"

```

16.12. state:New-Jersey (14 (0.1%))

Rule: Emitted as state:New-Jersey when '<place>, New Jersey' is captured and the state resolves against the U.S. state vocabulary.

```

264:      "New Jersey",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"

```

16.13. state:Colorado (13 (0.1%))

Rule: Emitted as state:Colorado when '<place>, Colorado' is captured and the state resolves against the U.S. state vocabulary.

```
240:      "Colorado",

1229: # "<Place>, <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> - the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.14. state:Oregon (13 (0.1%))

Rule: Emitted as state:Oregon when '<place>, Oregon' is captured and the state resolves against the U.S. state vocabulary.

```
271:      "Oregon",

1229: # "<Place>, <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> - the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.15. state:Pennsylvania (13 (0.1%))

Rule: Emitted as state:Pennsylvania when '<place>, Pennsylvania' is captured and the state resolves against the U.S. state vocabulary.

```
272:      "Pennsylvania",

1229: # "<Place>, <State>" - anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> - the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.16. state:Arizona (12 (0.1%))

Rule: Emitted as state:Arizona when '<place>, Arizona' is captured and the state resolves against the U.S. state vocabulary.

```

237:      "Arizona",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.17. state:Maryland (12 (0.1%))

Rule: Emitted as state:Maryland when '<place>, Maryland' is captured and the state resolves against the U.S. state vocabulary.

```

254:      "Maryland",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.18. state:Massachusetts (12 (0.1%))

Rule: Emitted as state:Massachusetts when '<place>, Massachusetts' is captured and the state resolves against the U.S. state vocabulary.

```

255:      "Massachusetts",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.19. state:Connecticut (9 (0.1%))

Rule: Emitted as state:Connecticut when '<place>, Connecticut' is captured and the state resolves against the U.S. state vocabulary.

```

241:      "Connecticut",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```


16.20. state:Georgia (9 (0.1%))

Rule: Emitted as state:Georgia when '<place>, Georgia' is captured and the state resolves against the U.S. state vocabulary.

```
244:      "Georgia",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.21. state:Louisiana (7 (0.1%))

Rule: Emitted as state:Louisiana when '<place>, Louisiana' is captured and the state resolves against the U.S. state vocabulary.

```
252:      "Louisiana",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.22. state:Michigan (7 (0.1%))

Rule: Emitted as state:Michigan when '<place>, Michigan' is captured and the state resolves against the U.S. state vocabulary.

```
256:      "Michigan",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.23. state:South-Carolina (7 (0.1%))

Rule: Emitted as state:South-Carolina when '<place>, South Carolina' is captured and the state resolves against the U.S. state vocabulary.

```

274:      "South Carolina",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.24. state:Missouri (6 (0.1%))

Rule: Emitted as state:Missouri when '<place>, Missouri' is captured and the state resolves against the U.S. state vocabulary.

```

259:      "Missouri",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.25. state:Nevada (6 (0.1%))

Rule: Emitted as state:Nevada when '<place>, Nevada' is captured and the state resolves against the U.S. state vocabulary.

```

262:      "Nevada",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.26. state:New-Hampshire (6 (0.1%))

Rule: Emitted as state:New-Hampshire when '<place>, New Hampshire' is captured and the state resolves against the U.S. state vocabulary.

```

263:      "New Hampshire",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.27. state:Wisconsin (6 (0.1%))

Rule: Emitted as state:Wisconsin when '<place>, Wisconsin' is captured and the state resolves against the U.S. state vocabulary.

```
283:      "Wisconsin",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.28. state:Alabama (5 (0.0%))

Rule: Emitted as state:Alabama when '<place>, Alabama' is captured and the state resolves against the U.S. state vocabulary.

```
235:      "Alabama",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.29. state:Idaho (5 (0.0%))

Rule: Emitted as state:Idaho when '<place>, Idaho' is captured and the state resolves against the U.S. state vocabulary.

```
246:      "Idaho",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.30. state:Iowa (5 (0.0%))

Rule: Emitted as state:Iowa when '<place>, Iowa' is captured and the state resolves against the U.S. state vocabulary.

```

249:      "Iowa",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.31. state:Vermont (5 (0.0%))

Rule: Emitted as state:Vermont when '<place>, Vermont' is captured and the state resolves against the U.S. state vocabulary.

```

279:      "Vermont",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.32. state:Indiana (4 (0.0%))

Rule: Emitted as state:Indiana when '<place>, Indiana' is captured and the state resolves against the U.S. state vocabulary.

```

248:      "Indiana",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.33. state:Kansas (4 (0.0%))

Rule: Emitted as state:Kansas when '<place>, Kansas' is captured and the state resolves against the U.S. state vocabulary.

```

250:      "Kansas",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.34. state:Mississippi (4 (0.0%))

Rule: Emitted as state:Mississippi when '<place>, Mississippi' is captured and the state resolves against the U.S. state vocabulary.

```
258:     "Mississippi",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.35. state:Kentucky (3 (0.0%))

Rule: Emitted as state:Kentucky when '<place>, Kentucky' is captured and the state resolves against the U.S. state vocabulary.

```
251:     "Kentucky",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.36. state:New-Mexico (3 (0.0%))

Rule: Emitted as state:New-Mexico when '<place>, New Mexico' is captured and the state resolves against the U.S. state vocabulary.

```
265:     "New Mexico",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703:     # state:<NAME> – the "<City>, <State>" pattern, validated.
1704:     for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:         resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:         if resolved:
1707:             add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046:     if low in ("usa", "united states"):
2047:         return "United-States"
2048:     if low == "uk":
2049:         return "United-Kingdom"
2050:     if low == "uae":
2051:         return "United-Arab-Emirates"
```

16.37. state:Oklahoma (3 (0.0%))

Rule: Emitted as state:Oklahoma when '<place>, Oklahoma' is captured and the state resolves against the U.S. state vocabulary.

```

270:      "Oklahoma",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.38. state:Hawaii (2 (0.0%))

Rule: Emitted as state:Hawaii when '<place>, Hawaii' is captured and the state resolves against the U.S. state vocabulary.

```

245:      "Hawaii",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.39. state:Ohio (2 (0.0%))

Rule: Emitted as state:Ohio when '<place>, Ohio' is captured and the state resolves against the U.S. state vocabulary.

```

269:      "Ohio",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.40. state:Alaska (1 (0.0%))

Rule: Emitted as state:Alaska when '<place>, Alaska' is captured and the state resolves against the U.S. state vocabulary.

```

236:      "Alaska",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.41. state:Arkansas (1 (0.0%))

Rule: Emitted as state:Arkansas when '<place>', Arkansas' is captured and the state resolves against the U.S. state vocabulary.

```
238:      "Arkansas",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
```

16.42. state:Delaware (1 (0.0%))

Rule: Emitted as state:Delaware when '<place>', Delaware' is captured and the state resolves against the U.S. state vocabulary.

```
242:      "Delaware",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
```

16.43. state:Maine (1 (0.0%))

Rule: Emitted as state:Maine when '<place>', Maine' is captured and the state resolves against the U.S. state vocabulary.

```
253:      "Maine",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"
```

16.44. state:Nebraska (1 (0.0%))

Rule: Emitted as state:Nebraska when '<place>', Nebraska' is captured and the state resolves against the U.S. state vocabulary.

```

261:     "Nebraska",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.45. state:North-Dakota (1 (0.0%))

Rule: Emitted as state:North-Dakota when '<place>, North Dakota' is captured and the state resolves against the U.S. state vocabulary.

```

268:     "North Dakota",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

16.46. state:South-Dakota (1 (0.0%))

Rule: Emitted as state:South-Dakota when '<place>, South Dakota' is captured and the state resolves against the U.S. state vocabulary.

```

275:     "South Dakota",

1229: # "<Place>, <State>" – anchors the STATE validator.
1230: PATTERN_STATE_CANDIDATE = re.compile(rf"\b{_PLACE},\s+({_PLACE})\b")

1703: # state:<NAME> – the "<City>, <State>" pattern, validated.
1704: for m in PATTERN_STATE_CANDIDATE.finditer(text):
1705:     resolved = _resolve_vocab(m.group(1) or "", STATE_BY_LOWER)
1706:     if resolved:
1707:         add(f"state:{_normalize_state(resolved)}", span=m.span(1))
1708:

2046: if low in ("usa", "united states"):
2047:     return "United-States"
2048: if low == "uk":
2049:     return "United-Kingdom"
2050: if low == "uae":
2051:     return "United-Arab-Emirates"

```

17. subject

The subject namespace is produced by direct subject regexes plus deterministic composites. subject:enforcement-op is added only when action:detention and theme:criminal have already both fired.

17.1. subject:enforcement-op (1,351 (13.5%))

Rule: Composite rule: action:detention and theme:criminal have both already fired.

```

1717: # subject:enforcement-op heuristic from the deterministic rules above:
1718: # if both action:detention and theme:criminal fire, that's strong enough
1719: # to set this without tentative.
1720: if any(e["tag"] == "action:detention" for e in entries) and any(
1721:     e["tag"] == "theme:criminal" for e in entries
1722: ):
1723:     add("subject:enforcement-op")
1724:
1725: # video:<kind> + video:duration-bucket – only when a video is attached.

```


17.2. subject:cbp-home-app (189 (1.9%))

Rule: Matches CBP Home App, CBP Home, DHS.GOV/CBPHOME, or DHS/CBP URLs containing cbp-home/cbphome/projecthomecoming.

```
780: PATTERN_SUBJECT_CBP_HOME_APP = _compile(
781:     r"\b(?:CBP\s+Home\s+App\b"
782:     r"|\b(?:CBP\s+Home\b"
783:     r"|\bDHS\.?GOV/CBPHOME\b"
784:     r"|\b(?:dhs|cbp)\.gov/(?:cbp[-_]?home|projecthomecoming)\b"
785: )
1544: (PATTERN_SUBJECT_CBP_HOME_APP, "subject:cbp-home-app"),
```

17.3. subject:angel-family (54 (0.5%))

Rule: Matches angel family/families/mom/dad/parent/mother/father/wife/husband/son/daughter/child.

```
1209: PATTERN_ANGEL_FAMILY = _compile(
1210:     r"\bangel(?:famil(?:y|ies)|mom|dad|parent|mother|father|wife|husband|son|daughter|child)\b"
1211: )
1590: (PATTERN_ANGEL_FAMILY, "subject:angel-family"),
```

17.4. subject:native-born-citizen (3 (0.0%))

Rule: Matches native-born citizens/Americans/U.S. citizens/people/workers/taxpayers.

```
1212: PATTERN_NATIVE_BORN_CITIZEN = _compile(
1213:     r"\bnative[- ]born\s+(?:citizens?|americans?|u\.?s\.?s+citizens?|people|workers|taxpayers)\b"
1214: )
1591: (PATTERN_NATIVE_BORN_CITIZEN, "subject:native-born-citizen"),
```

17.5. subject:celebrity (2 (0.0%))

Rule: Matches Sydney Sweeney, a celebrity-title phrase followed by a capitalized full name, or a capitalized full name followed by actress/influencer/celebrity/pop star/movie star.

```
799: PATTERN_SUBJECT_CEBLEBRITY = _compile(
800:     r"\bSydney\s+Sweeney\b"
801:     r"|\b(?:actress|influencer|celebrity|pop\s+star|movie\s+star|Hollywood\s+star)"
802:     r"\s+[A-Z][a-z]+(?:\s+[A-Z][a-z]+)+\b"
803:     r"|\b[A-Z][a-z]+(?:\s+[A-Z][a-z]+)+,\s+(?:an?s+)"
804:     r"|(?:actress|influencer|celebrity|pop\s+star|movie\s+star)\b"
805: )
1545: (PATTERN_SUBJECT_CEBLEBRITY, "subject:celebrity"),
```

17.6. subject:detainee (1 (0.0%))

Rule: No deterministic lexical emitter in scripts/tag_lexical.py; this appears only if imported from a vetted sidecar/manual tag source and normalized.

```

1364:     "media:short-video": ("video:short",),
1365:     # Namespace-migration aliases (media:* status flags -> media-status:*)
1366:     "media:archived": ("media-status:archived",),
1367:     "media:described": ("media-status:described",),
1368:     "media:has-alt-text": ("media-status:has-alt-text",),
1369:     "media:needs-vision": ("media-status:needs-vision",),
1370:     "media:needs-ocr": ("media-status:needs-ocr",),
1371:     "media:graphic-content": ("media-status:graphic-content",),
1372:     # Namespace-migration aliases (theme:/frame: -> policy:/format:/theme:)
1373:     "theme:border": ("policy:border",),
1374:     "theme:sanctuary-cities": ("policy:sanctuary-cities",),
1375:     "theme:worksite-enforcement": ("policy:worksite-enforcement",),
1376:     "theme:cbp-home": ("policy:cbp-home",),
1377:     "theme:statistics": ("format:statistics",),
1378:     "format:directive": ("theme:directive",),
1379:     "frame:criminal": ("theme:criminal",),
1380: }
1381:
1382:
1383: # -----
1384: # Per-tweet tagger.
1385: # -----
1386:
1387:
1388: def normalized_media_tags(media_tags: list[Any] | None) -> list[dict[str, Any]]:
1389:     """Return vetted tag entries imported from media-recognition sidecars."""
1390:     out: list[dict[str, Any]] = []
1391:     seen: set[str] = set()
1392:     if not isinstance(media_tags, list):
1393:         return out
1394:     for entry in media_tags:
1395:         if isinstance(entry, str):
1396:             tag = entry.strip()
1397:             source = "media-description"

```

```

1398:             tentative = False
1399:             elif isinstance(entry, dict):
1400:                 tag = str(entry.get("tag") or "").strip()
1401:                 source = str(entry.get("source") or "media-description")
1402:                 tentative = bool(entry.get("tentative"))
1403:             else:
1404:                 continue
1405:             if not tag or tag in seen:
1406:                 continue
1407:             aliased_tags = LEGACY_MEDIA_TAG_ALIASES.get(tag, (tag,))
1408:             for aliased_tag in aliased_tags:
1409:                 if aliased_tag in seen:
1410:                     continue
1411:                 if not any(
1412:                     aliased_tag.startswith(prefix) for prefix in MEDIA_TAG_PREFIXES_ALLOWED_IN_LEXICAL
1413:                 ):
1414:                     continue
1415:                 seen.add(aliased_tag)

```

This tag has no hard-coded lexical emitter in scripts/tag_lexical.py; the code above imports vetted sidecar/manual media tags when present.

17.7. subject:official (1 (0.0%))

Rule: No deterministic lexical emitter in scripts/tag_lexical.py; this appears only if imported from a vetted sidecar/manual tag source and normalized.

```

1364:     "media:short-video": ("video:short",),
1365:     # Namespace-migration aliases (media:* status flags -> media-status:*)
1366:     "media:archived": ("media-status:archived",),
1367:     "media:described": ("media-status:described",),
1368:     "media:has-alt-text": ("media-status:has-alt-text",),
1369:     "media:needs-vision": ("media-status:needs-vision",),
1370:     "media:needs-ocr": ("media-status:needs-ocr",),
1371:     "media:graphic-content": ("media-status:graphic-content",),
1372:     # Namespace-migration aliases (theme:/frame: -> policy:/format:/theme:)
1373:     "theme:border": ("policy:border",),
1374:     "theme:sanctuary-cities": ("policy:sanctuary-cities",),
1375:     "theme:worksite-enforcement": ("policy:worksite-enforcement",),
1376:     "theme:cbp-home": ("policy:cbp-home",),
1377:     "theme:statistics": ("format:statistics",),
1378:     "format:directive": ("theme:directive",),
1379:     "frame:criminal": ("theme:criminal",),
1380: }
1381:
1382:
1383: # -----
1384: # Per-tweet tagger.
1385: # -----
1386:
1387:
1388: def normalized_media_tags(media_tags: list[Any] | None) -> list[dict[str, Any]]:
1389:     """Return vetted tag entries imported from media-recognition sidecars."""
1390:     out: list[dict[str, Any]] = []
1391:     seen: set[str] = set()
1392:     if not isinstance(media_tags, list):
1393:         return out
1394:     for entry in media_tags:
1395:         if isinstance(entry, str):
1396:             tag = entry.strip()
1397:             source = "media-description"

```

```

1398:         tentative = False
1399:         elif isinstance(entry, dict):
1400:             tag = str(entry.get("tag") or "").strip()
1401:             source = str(entry.get("source") or "media-description")
1402:             tentative = bool(entry.get("tentative"))
1403:         else:
1404:             continue
1405:         if not tag or tag in seen:
1406:             continue
1407:         aliased_tags = LEGACY_MEDIA_TAG_ALIASES.get(tag, (tag,))
1408:         for aliased_tag in aliased_tags:
1409:             if aliased_tag in seen:
1410:                 continue
1411:             if not any(
1412:                 aliased_tag.startswith(prefix) for prefix in MEDIA_TAG_PREFIXES_ALLOWED_IN_LEXICAL
1413:             ):
1414:                 continue
1415:             seen.add(aliased_tag)

```

This tag has no hard-coded lexical emitter in scripts/tag_lexical.py; the code above imports vetted sidecar/manual media tags when present.

18. theme

The theme namespace is produced by rhetorical-frame regexes and guard functions. Those helpers handle expletive suppression for religion, context windows for coded nativism, and context windows for martyrdom.

18.1. theme:criminal (2,782 (27.8%))

Rule: Matches criminal illegal alien(s), illegal alien(s), criminal alien(s), aggravated felon(s), or convicted for/of.

```

515: PATTERN_FRAME_CRIMINAL = _compile(
516:     r"\b(criminal illegal aliens?|illegal aliens?|criminal aliens?|aggravated felons?|convicted (?for|of))\b"
517: )

1539:     (PATTERN_FRAME_CRIMINAL, "theme:criminal"),

```

18.2. theme:directive (420 (4.2%))

Rule: Matches sentence-start imperatives: Apply, Report, Call, Visit, Leave, Self-deport, Go to, Submit, Sign up, Tip, Click, Tap, See, or Read.

```

1206: PATTERN_THEME_DIRECTIVE = _compile(
1207:     r"(?:^[!?\s+](Apply|Report|Call|Visit|Leave|Self[- ]deport|Go to|Submit|Sign up|Tip|Click|Tap|See|Read)\b"
1208: )

1589:     (PATTERN_THEME_DIRECTIVE, "theme:directive"),

```

18.3. theme:religion (329 (3.3%))

Rule: Matches religion/religious, faith/faithful/faith-based, worship/church/synagogue/mosque/temple/chaplain, prayer/pray/praying/prayed, bless/blessed/blessing, God/Lord/Jesus/Christ/Christian, Bible/biblical/scripture. Suppresses common expletives within 18 characters.

```

697: PATTERN_THEME_RELIGION = _compile(
698:     r"\breligio(?:n|us)\b"
699:     r"|\bfaith(?:ful|s|[- ]based)?\b"
700:     r"|\b(?:worship|church(?:es)?|synagogue|mosque|temple|chaplain)\b"
701:     r"|\b(?:prayer|prayers|pray|praying|prayed)\b"
702:     r"|\b(?:bless|blessed|blessing|blessings)\b"
703:     r"|\b(?:god|lord|jesus|christ|christian(?:ity|s)?|judeo[- ]christian)\b"
704:     r"|\b(?:bible|biblical|scripture|scriptural)\b"
705: )

706: PATTERN_THEME_RELIGION_EXPLETIVE = _compile(
707:     r"\b(?:oh\s+my\s+god|omg|god\s+damn(?:ed|it)?|goddamn(?:ed)?|"
708:     r"for\s+god'?s\s+sake|good\s+lord|thank\s+god)\b"
709: )

1777: for match in PATTERN_THEME_RELIGION.finditer(text):
1778:     start, end = match.span()
1779:     window = text[
1780:         max(0, start - RELIGION_EXPLETIVE_WINDOW_CHARS) : min(
1781:             len(text), end + RELIGION_EXPLETIVE_WINDOW_CHARS
1782:         )
1783:     ]
1784:     if PATTERN_THEME_RELIGION_EXPLETIVE.search(window):
1785:         continue
1786:     return match
1787: return None

```

18.4. theme:martyrdom (202 (2.0%))

Rule: Matches Angel Family/family-member language; memorial/victim/fallen/killed/murdered language within 180 characters of honor/remember/commemorate/mourn/never forget/say their names in either order; would still be alive/life was taken/preventable

tragedy/line of duty/gave everything. This is our why also matches directly for tracked accounts.

```
770: PATTERN_MARTYRDOM_CONTEXT = _compile(
771:     r"\bangel\s+(?:famil(?:y|ies)|mom|dad|parent|mother|father|wife|husband|son|daughter|child)\b"
772:     r"|\b(?:honou?r|remember|commemorate|mourn|never\s+forget|say\s+their\s+names?)\b"
773:     r".{0,180}\b(?:victims?|fallen|killed|murdered|lives?|life|names?|memory|heroes|families)\b"
774:     r"|\b(?:victims?|fallen|killed|murdered|lives?|life|names?|memory|heroes|families)\b"
775:     r".{0,180}\b(?:honou?r|remember|commemorate|mourn|never\s+forget|say\s+their\s+names?)\b"
776:     r"|\b(?:would\s+still\s+be\s+alive|life\s+was\s+taken|lives?\s+were\s+taken|"
777:     r"preventable\s+(?:murder|death|tragedy)|killed\s+in\s+the\s+line\s+of\s+duty|"
778:     r"gave\s+everything)\b"
779: )

765: PATTERN_MARTYRDOM_WHY = _compile(
766:     r"\bthis\s+is\s+our\s+why\b"
767:     r"|\bthey\s+are\s+our\s+why\b"
768:     r"|\byou\s+are\s+why\s+we\s+fight\b"
769: )

1831: if m := PATTERN_MARTYRDOM_CONTEXT.search(text):
1832:     return m
1833:
1834: why = PATTERN_MARTYRDOM_WHY.search(text)
1835: if not why:
1836:     return None
1837: existing = {str(e["tag"]) for e in entries}
1838: if account_category in {"core", "government", "officials"}:
1839:     return why
1840: if existing.intersection({"subject:angel-family", "subject:crime-victim", "crime:homicide"}):
1841:     return why
1842: start, end = why.span()
1843: window = text[max(0, start - 220) : min(len(text), end + 220)]
1844: if PATTERN_MARTYRDOM_CONTEXT.search(window):
1845:     return why
```

18.5. theme:civil-disturbance (85 (0.8%))

Rule: Matches riot/rioters/rioting, civil disturbance/unrest, mass unrest, violent protest(s), anti-ICE riot/protest, violent demonstrators, street violence, or mob violence.

```
725: PATTERN_THEME_CIVIL_DISTURBANCE = _compile(
726:     r"\b(?:riot(?:s|ers|ing)?|civil\s+disturbance|civil\s+unrest|mass\s+unrest|"
727:     r"violent\s+protests?|anti[- ]ICE\s+(?:riot(?:s|ers|ing)?|protests?)|"
728:     r"violent\s+demonstrators?|street\s+violence|mob\s+violence)\b"
729: )

1559: (PATTERN_THEME_CIVIL_DISTURBANCE, "theme:civil-disturbance"),
```

18.6. theme:homeland (67 (0.7%))

Rule: Matches our/the/this/America's Homeland; protect/secure/defend/safeguard Homeland; or safe/secure Homeland.

```
657: PATTERN_THEME_HOMELAND = re.compile(
658:     r"\b(?:our|the|this|america's)\s+Homeland\b"
659:     r"|\b(?:protect|protecting|secure|securing|defend|defending|safeguard|safeguarding)"
660:     r"s+(?:(?:our|the|this|america's)\s+)?Homeland\b"
661:     r"|\b(?:safe|secure)\s+Homeland\b"
662: )

1555: (PATTERN_THEME_HOMELAND, "theme:homeland"),
```

18.7. theme:nativism (30 (0.3%))

Rule: Matches native-born/American-born people, foreign-born/foreign workers with displacement/flood/cheap/replacement/betrayal/American-worker context within 140 characters, globalism has failed/Americanism will prevail, forefathers, birthright-as-stolen language, or inheritance language within 160 characters of nation/citizenship/Homeland/birthright/immigration context.

```

663: PATTERN_THEME_NATIVISM = _compile(
664:     r"\bnative[- ]born\s+(?:americans?|workers?|citizens?)\b"
665:     r"|\bamerican[- ]born\s+(?:americans?|workers?|citizens?)\b"
666:     r"|\bforeign[- ]born\s+workers?\b"
667:     r"|\bforeign\s+(?:workers?|labor)\b.{0,140}\b(?:flood|cheap|displac|replac|"
668:     r"betray|job market|american\s+(?:workers?|jobs?)|americans?\s+first)\b"
669:     r"|\b(?:american\s+(?:workers?|jobs?)|americans?\s+first|job market)\b.{0,140}"
670:     r"\b(?:foreign\s+(?:workers?|labor)|foreign[- ]born\s+workers?)\b"
671:     r"|\bglobalism has failed\b|bamericanism will prevail\b"
672:     r"|\b(?:our\s+)?forefathers?\b"
673: )

821: PATTERN_THEME_NATIVISM_BIRTHRIGHT = _compile(
822:     r"\b(?:actual|real|true)\s+birthright\s+of\s+(?:every|all)?\s*americans?\b"
823:     r"|\bamerican(?:'s)?\s+birthright\b"
824:     r"|\b(?:our|the)\s+birthright\b.{0,160}\b(?:steal|steals?|stolen|dilut|eras|destroy|destroyed|replac|betray|undermin)\b"
825:     r"|\b(?:steal|steals?|stolen|dilut|eras|destroy|destroyed|replac|betray|undermin)\b.{0,160}\b(?:our|the)\s+birthright\b"
826:     r"|\b(?:destroy(?:ing)?|destroys?|erasing?|replac(?:e|ing)?|betray(?:ing)?|undermin(?:e|ing)?|dilut(?:e|ing)?)"
827:     r"\b.{0,120}\b(?:our|the|american(?:'s)?)\s+birthright\b"
828: )

1809:         "topic:immigration",
1810:     }
1811:     for tag in existing
1812:     )
1813:     for match in PATTERN_NATIVISM_INHERITANCE.finditer(text):
1814:         start, end = match.span()
1815:         window = text[max(0, start - 160) : min(len(text), end + 160)]
1816:         if strong_tag_context or PATTERN_NATIVISM_INHERITANCE_CONTEXT.search(window):
1817:             return match
1818:     return None
1819:

```

18.8. theme:transgender (23 (0.2%))

Rule: Matches transgender, gender ideology/identity/affirming/transition/dysphoria, biological/transgender male/female/man/woman/boy/girl language, men/boys in women's/girls' sports, women/girls sports exclusion/protection language, or Title IX within 80 characters of transgender/gender/sports context.

```

711: PATTERN_THEME_TRANSGENDER = _compile(
712:     r"\btransgender\b"
713:     r"|\bgender[- ](?:ideology|identity|affirming|transition|dysphoria)\b"
714:     r"|\b(?:biological|transgender)\s+"
715:     r"|(?:males?|females?|man|woman|men|women|boys?|girls?)\b"
716:     r"|\b(?:men|males|boys)\s+in\s+(?:women[\u2019]?s|girls[\u2019]?)\s+sports\b"
717:     r"|\b(?:men|males|boys)\s+(?:competing|playing|participating)\s+"
718:     r"|(?:in|against|with)\s+(?:women|girls|female\s+athletes?)\b"
719:     r"|\b(?:women[\u2019]?s|girls[\u2019]?)\s+sports\s+(?:are\s+)\b"
720:     r"for\s+(?:women|girls|female\s+athletes?)\b"
721:     r"|\b(?:protect|save|defend)\s+(?:women[\u2019]?s|girls[\u2019]?)\s+sports\b"
722:     r"|\btitle\s+ix\b.{0,80}\b(?:transgender|gender|women[\u2019]?s\s+sports|girls[\u2019]?s\s+sports)\b"
723:     r"|\b(?:transgender|gender|women[\u2019]?s\s+sports|girls[\u2019]?s\s+sports)\b.{0,80}\btitle\s+ix\b"
724: )

1558:         (PATTERN_THEME_TRANSGENDER, "theme:transgender"),

```

18.9. theme:pop-culture-reference (5 (0.0%))

Rule: Matches Sydney Sweeney; American Eagle within 100 characters of ICE/DHS/CBP/deport/illegal alien/border in either order; or good genes within 100 characters of jeans/ICE/DHS/CBP/deport/border/illegal alien. Also implied by parody matches.

```

806: PATTERN_THEME_POP_CULTURE_REFERENCE = _compile(
807:     r"\bSydney\s+Sweeney\b"
808:     r"|\bAmerican\s+Eagle\b.{0,100}\b(?:ICE|DHS|CBP|deport|illegal\s+alien|border)\b"
809:     r"|\b(?:ICE|DHS|CBP|deport|illegal\s+alien|border)\b.{0,100}\bAmerican\s+Eagle\b"
810:     r"|\bgood\s+genes\b.{0,100}\b(?:jeans|ICE|DHS|CBP|deport|border|illegal\s+alien)\b"
811: )

1561:         (PATTERN_THEME_POP_CULTURE_REFERENCE, "theme:pop-culture-reference"),

```

19. topic

The topic namespace is produced by direct topic regexes, parent-topic implications from narrower tags, and the immigration sticky default. For tracked core/government/official accounts, topic:immigration is added unless a non-immigration blocker fires; it is tentative when no explicit immigration signal exists.

19.1. topic:immigration (9,196 (91.9%))

Rule: Added when explicit immigration regex matches immigration/immigrant/migrant/asylum/illegal alien/undocumented alien/border patrol/CBP Home/deport/removal; or when narrower immigration tags imply it; or by default for core/government/official accounts unless a weather/disaster/sports/birthday/anniversary blocker or another non-immigration topic prevents it. Tentative when only the account default supports it.

```

1908: PATTERN_EXPLICIT_IMMIGRATION_TOPIC = _compile(
1909:     r"\b(immigration|immigrants?|migrants?|asylum|illegal\s+(?:alien|immigrant)s?)|"
1910:     r"undocumented\s+(?:alien|immigrant)s?|border patrol|CBP\s+Home|"
1911:     r"deport(?:ed|ing|ation)?|removals?)\b"
1912: )

1855: "action:deportation": ("topic:immigration",),
1856: "action:self-deportation": ("topic:immigration",),
1857: "action:report-immigrants": ("topic:immigration",),
1858: "agency:ICEgov": ("topic:immigration",),
1859: "agency:CBP": ("topic:immigration",),
1860: "agency:DHSgov": ("topic:immigration",),
1861: "agency:HSI_HQ": ("topic:immigration",),
1862: "agency:USBPChief": ("topic:immigration",),
1863: "agency:DeptofWar": ("topic:military",),
1864: "agency:CENTCOM": ("topic:military",),
1865: "agency:Southcom": ("topic:military",),
1866: "agency:USCG": ("topic:military",),
1867: "agency:USArmyNorth": ("topic:military",),
1868: "agency:USNationalGuard": ("topic:military",),
1869: "agency:USNorthernCmd": ("topic:military",),
1870: "theme:criminal": ("topic:immigration",),
1871: "genre:lineup": ("topic:immigration",),
1872: "subject:angel-family": ("theme:martyrdom", "topic:immigration"),
1873: "subject:crime-victim": ("theme:martyrdom", "topic:immigration"),
1874: "subject:cbp-home-app": ("topic:immigration",),
1875: "subject:enforcement-op": ("topic:immigration",),
1876: "policy:border": ("topic:immigration",),
1877: "policy:cbp-home": ("topic:immigration",),
1878: "policy:sanctuary-cities": ("topic:immigration",),
1879: "policy:worksite-enforcement": ("topic:economy", "topic:immigration"),
1880: "theme:nativism": ("topic:immigration",),
1881: "theme:pop-culture-reference": ("topic:immigration",),
1882: "religion:christianity": ("theme:religion",),

1883: "slogan:criminal-illegal-alien": ("topic:immigration",),
1884: "slogan:free-ticket-home": ("topic:immigration",),
1885: "slogan:go-home": ("topic:immigration",),
1886: "slogan:illegal-alien": ("topic:immigration",),
1887: "slogan:mass-deportation": ("topic:immigration",),
1888: "slogan:masa": ("topic:immigration",),
1889: "slogan:find-and-kill": ("topic:immigration",),
1890: "slogan:import-third-world": ("topic:immigration",),
1891: "legal:birthright-citizenship": ("topic:immigration",),
1892: "genre:parody": ("theme:pop-culture-reference",),
1893: "slogan:most-secure-border": ("topic:immigration", "policy:border"),
1894: "slogan:catch-release": ("topic:immigration",),
1895: "slogan:project-homecoming": ("topic:immigration",),
1896: "slogan:maga": ("topic:general",),
1897: "slogan:maha": ("topic:general",),
1898: "slogan:america-first": ("topic:general",),
1899: "slogan:golden-age": ("topic:general",),
1900: "slogan:save-america": ("topic:general",),
1901: "slogan:law-and-order": ("topic:general",),
1902: "slogan:peace-through-strength": ("topic:general",),
1903: "slogan:promises-kept": ("topic:general",),
1904: "phrase:immigrant": ("topic:immigration",),
1905: "phrase:migrant": ("topic:immigration",),
1906: }
1907: INTRINSIC_PARENT_TOPICS_PREFIXES: tuple[tuple[str, str], ...] = (("origin:", "topic:immigration"),)
1908: PATTERN_EXPLICIT_IMMIGRATION_TOPIC = _compile(
1909:     r"\b(immigration|immigrants?|migrants?|asylum|illegal\s+(?:alien|immigrant)s?)|"

1936: if PATTERN_EXPLICIT_IMMIGRATION_TOPIC.search(text):
1937:     add("topic:immigration")
1938: if (
1939:     any(e["tag"] == "policy:worksite-enforcement" for e in entries)
1940:     and "topic:economy" not in existing_tags
1941: ):

```

```

1942:         add("topic:economy")
1943:
1944:
1945: # Tag names whose presence on a tweet promotes `topic:immigration` from
1946: # a tentative-by-account default to a confirmed-by-text classification.
1947: # Anything that explicitly references the immigration domain (origin
1948: # country pattern, deportation verb, border keyword, ICE/CBP/DHS handle,
1949: # the criminal-alien frame, etc.) clears the bar.
1950: IMMIGRATION_CONFIRMING_PREFIXES: tuple[str, ...] = (
1951:     "action:",
1952:     "origin:",
1953:     "country:",
1954:     "policy:border",
1955:     "policy:sanctuary",
1956:     "policy:worksite",
1957:     "policy:cbp-home",
1958:     "theme:nativism",
1959:     "theme:criminal",
1960:     "shape:",
1961:     "subject:cbp-home-app",
1962:     "subject:enforcement-op",
1963: )
1964: IMMIGRATION_CONFIRMING_EXACT: frozenset[str] = frozenset(
1965:     {
1966:         "agency:ICEgov",
1967:         "agency:CBP",
1968:         "agency:DHSgov",
1969:         "agency:HSI_HQ",
1970:         "agency:USBPCchief",
1971:         "slogan:criminal-illegal-alien",
1972:         "slogan:free-ticket-home",
1973:         "slogan:go-home",
1974:         "slogan:illegal-alien",
1975:         "slogan:mass-deportation",
1976:         "slogan:masa",
1977:         "slogan:most-secure-border",
1978:         "slogan:catch-release",
1979:         "slogan:nice",
1980:         "slogan:project-homecoming",
1981:         "slogan:reportrecon",
1982:         "slogan:worst",
1983:         "phrase:immigrant",
1984:         "phrase:migrant",
1985:     }
1986: )
1987: # Last-ditch keyword check – picks up image-heavy / template-light
1988: # tweets that the namespace rules above don't flag but that still
1989: # clearly read as immigration content ("ICE arrested", "illegal alien",
1990: # "the border", standalone "immigration"). Kept narrow on purpose.
1991: PATTERN_IMMIGRATION_KEYWORD = _compile(
1992:     r"\b(immigration|immigrant|migrant|asylum|illegal alien|the border|"
1993:     r"border patrol|ICE\b|CBP\b)\b"
1994: )
1995:
1996:
1997: def _has_immigration_signal(entries: list[dict[str, Any]], text: str) -> bool:
1998:     """True if the tweet carries any explicit immigration-domain signal.
1999:     Drives the confirmed-vs-tentative split for `topic:immigration`."""
2000:     for e in entries:
2001:         tag = e["tag"]
2002:         if tag in IMMIGRATION_CONFIRMING_EXACT:
2003:             return True
2004:         for pref in IMMIGRATION_CONFIRMING_PREFIXES:
2005:             if tag.startswith(pref):
2006:                 return True
2007:     return bool(PATTERN_IMMIGRATION_KEYWORD.search(text))
2008:
2009:

```

```

2010: def _has_non_immigration_topic(entries: list[dict[str, Any]]) -> bool:
2011:     return any(e["tag"].startswith("topic:") and e["tag"] != "topic:immigration" for e in entries)
2012:
2013:
2014: def _maybe_immigration_default(
2015:     entries: list[dict[str, Any]],
2016:     account_category: str,
2017:     text: str,
2018:     add: Callable[..., None],
2019: ) -> None:
2020:     """Apply `topic:immigration` to every tweet from a tracked-tier
2021:     author unless an obvious non-immigration signal blocks it.
2022:     Confirmed when the tweet carries any explicit immigration-domain
2023:     marker; tentative otherwise (image-heavy, template-light tweets
2024:     where the author identity is the only signal we have).
2025:
2026:     See the `topic:immigration` entry in `config/tag_taxonomy.yaml` for
2027:     the rationale."""
2028:     if account_category not in IMMIGRATION_DEFAULT_CATEGORIES:
2029:         return
2030:     has_signal = _has_immigration_signal(entries, text)
2031:     if has_signal:
2032:         add("topic:immigration")
2033:         return
2034:     if any(e["tag"] == "event:palestine" for e in entries):
2035:         return
2036:     for pat in NON_IMMIGRATION_PATTERNS:
2037:         if pat.search(text):
2038:             return
2039:     if not _has_non_immigration_topic(entries):
2040:         add("topic:immigration", tentative=True)
2041:

```

19.2. topic:economy (1,083 (10.8%))

Rule: Matches economy/economic, jobs/job growth, workers/workforce/wages/labor market, employment/unemployment/hiring/manufacturing/business/apprenticeship/small business, taxes/tax cuts, inflation/prices/cost of living, tariffs/trade deficit/supply chain/GDP/stock market/markets. Also implied by policy:worksite-enforcement.

```

560: PATTERN_TOPIC_ECONOMY = _compile(
561:     r"\b(econom(?:y|ic)|jobs?|job growth|workers?|workforce|wages?|labor market|"
562:     r"employment|unemployment|hiring|manufactur(?:e|ing)|business(?:es)?|"
563:     r"apprenticeships?|small businesses?|tax cuts?|taxes|inflation|prices?|"
564:     r"cost of living|tariffs?|trade deficit|supply chains?|GDP|stock market|markets?)\b"
565: )
1548: (PATTERN_TOPIC_ECONOMY, "topic:economy"),

```

19.3. topic:general (971 (9.7%))

Rule: Added when at least three broad problem-domain regexes match among immigration, crime, economy, military, fraud, security, and costs; also implied by broad slogans such as MAGA/MAHA/America First/Golden Age/Save America/Law and Order/Peace Through Strength/Promises Kept.

```

643: GENERAL_TOPIC_PATTERNS: tuple[tuple[str, re.Pattern[str]], ...] = (
644:     ("immigration", _compile(r"\b(immigration|migrant|border|illegal alien|asylum)\b")),
645:     ("crime", _compile(r"\b(crime|criminal|drugs?|fentanyl|gang|violence|murder)\b")),
646:     ("economy", PATTERN_TOPIC_ECONOMY),
647:     ("military", PATTERN_TOPIC_MILITARY),
648:     ("fraud", _compile(r"\b(fraud|waste|abuse|corruption|scam)\b")),
649:     ("security", _compile(r"\b(security|terror|war|china|cartel|threat)\b")),
650:     ("costs", _compile(r"\b(inflation|prices?|taxes|cost of living)\b")),
651: )
1848:
1849: def _general_topic_score(text: str) -> int:
1850:     """Count broad problem domains in a multi-issue / grievance-style post."""
1851:     return sum(1 for _slug, pat in GENERAL_TOPIC_PATTERNS if pat.search(text))
1852:
1909: r"\b(immigration|immigrants?|migrants?|asylum|illegal\s+(?:alien|immigrant)s?|"
1910: r"undocumented\s+(?:alien|immigrant)s?|border patrol|CBP\s+Home|"
1911: r"deport(?:ed|ing|ation)?|removals?)\b"
1912: )
1913:
1914:
1915: def _ensure_intrinsic_parent_topics(
1916:     entries: list[dict[str, Any]],
1917:     text: str,
1918:     add: Callable[..., None],
1919: ) -> None:
1920:     """Add broad topic parents implied by narrower deterministic tags."""
1921:     existing_tags = {str(e["tag"]) for e in entries}
1922:     if (
1923:         any(tag.startswith("military:") for tag in existing_tags)
1924:         and "topic:military" not in existing_tags
1925:     ):
1926:         add("topic:military")

```



```

1927:     existing_tags.add("topic:military")
1928:     for tag in list(existing_tags):
1929:         for parent in INTRINSIC_PARENT_TOPICS_EXACT.get(tag, ()):
1930:             add(parent)
1931:     existing_tags.add(parent)
1932:     for prefix, parent in INTRINSIC_PARENT_TOPICS_PREFIXES:
1933:         if tag.startswith(prefix):
1934:             add(parent)

```

19.4. topic:military (658 (6.6%))

Rule: Matches military/armed forces/service members/veterans/troops/soldiers/sailors/airmen/marines/guardsmen, service branches, Pentagon/DoD/VA, combat/battlefield/war zone/deployment, carrier/USS/USNS terms, service academies/cadets/midshipmen, commander-in-chief, or commandant. Also implied by any military:<branch> tag.

```

566: PATTERN_TOPIC_MILITARY = _compile(
567:     r"\b(military|armed forces|servicemembers?|service members?|veterans?|troops?|"
568:     r"soldiers?|sailors?|airmen|marines|guardsmen|army|navy|air force|space force|"
569:     r"marine corps|coast guard|national guard|USAF|USSF|USMC|pentagon|"
570:     r"department of defense|DoD|DOD|veterans affairs|combat|battlefield|"
571:     r"war zone|deployed|deployment|carrier strike group|aircraft carrier|"
572:     r"carrier air wing|CVN\s*\d+|USS\s+[A-Z][A-Za-z0-9-]+|USNS\s+[A-Z][A-Za-z0-9-]+|"
573:     r"service academ(?:y|ies)|military academy|naval academy|coast guard academy|"
574:     r"air force academy|west point|USMA|USNA|USCGA|cadets?|midshipmen|"
575:     r"commander[- ]in[- ]chief|commandant\b"
576: )
1549:     (PATTERN_TOPIC_MILITARY, "topic:military"),

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19.5. topic:laudatory (99 (1.0%))

Rule: Matches accomplishment-list framing: Promises Made/Kept, Golden Age of America, two or more check-mark bullets, delivering on promises/for Americans, historic/record/biggest/greatest/unprecedented/largest accomplishment/achievement/win/progress/result/deal, list/litany/series of accomplishments, or administration/president/Trump accomplishment framing.

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621: PATTERN_TOPIC_LAUDATORY = _compile(
622:     # Laudatory = a general touting of the administration's accomplishments
623:     # (a "laundry list" of wins), NOT any single booster word. Bare words like
624:     # "historic", "delivering", "winning", "momentum", "success" produced heavy
625:     # false positives ("delivering remarks", "Massie winning", "vaccine
626:     # delivered", "historic astronauts"), so require explicit accomplishment-
627:     # list framing: signature slogans, check-marked bullet lists, or an
628:     # accomplishment noun qualified as historic/record/biggest/etc.
629:     r"\bpromises?s+made[,;:.]?s+promises?s+kept\b"
630:     r"\bgolden\s+age\s+of\s+america\b"
631:     r"|(?:[\u2022\u2023\u2024\u2025\u2026\u2027\u2028\u2029\u2030\u2031\u2032\u2033\u2034\u2035\u2036\u2037\u2038\u2039\u203A\u203B\u203C\u203D\u203E\u203F\u2040\u2041\u2042\u2043\u2044\u2045\u2046\u2047\u2048\u2049\u204A\u204B\u204C\u204D\u204E\u204F\u2050\u2051\u2052\u2053\u2054\u2055\u2056\u2057\u2058\u2059\u205A\u205B\u205C\u205D\u205E\u205F\u2060\u2061\u2062\u2063\u2064\u2065\u2066\u2067\u2068\u2069\u206A\u206B\u206C\u206D\u206E\u206F\u2070\u2071\u2072\u2073\u2074\u2075\u2076\u2077\u2078\u2079\u207A\u207B\u207C\u207D\u207E\u207F\u2080\u2081\u2082\u2083\u2084\u2085\u2086\u2087\u2088\u2089\u208A\u208B\u208C\u208D\u208E\u208F\u2090\u2091\u2092\u2093\u2094\u2095\u2096\u2097\u2098\u2099\u209A\u209B\u209C\u209D\u209E\u209F\u2100\u2101\u2102\u2103\u2104\u2105\u2106\u2107\u2108\u2109\u210A\u210B\u210C\u210D\u210E\u210F\u2110\u2111\u2112\u2113\u2114\u2115\u2116\u2117\u2118\u2119\u211A\u211B\u211C\u211D\u211E\u211F\u2120\u2121\u2122\u2123\u2124\u2125\u2126\u2127\u2128\u2129\u212A\u212B\u212C\u212D\u212E\u212F\u2130\u2131\u2132\u2133\u2134\u2135\u2136\u2137\u2138\u2139\u213A\u213B\u213C\u213D\u213E\u213F\u2140\u2141\u2142\u2143\u2144\u2145\u2146\u2147\u2148\u2149\u214A\u214B\u214C\u214D\u214E\u214F\u2150\u2151\u2152\u2153\u2154\u2155\u2156\u2157\u2158\u2159\u215A\u215B\u215C\u215D\u215E\u215F\u2160\u2161\u2162\u2163\u2164\u2165\u2166\u2167\u2168\u2169\u216A\u216B\u216C\u216D\u216E\u216F\u2170\u2171\u2172\u2173\u2174\u2175\u2176\u2177\u2178\u2179\u217A\u217B\u217C\u217D\u217E\u217F\u2180\u2181\u2182\u2183\u2184\u2185\u2186\u2187\u2188\u2189\u218A\u218B\u218C\u218D\u218E\u218F\u2190\u2191\u2192\u2193\u2194\u2195\u2196\u2197\u2198\u2199\u219A\u219B\u219C\u219D\u219E\u219F\u21A0\u21A1\u21A2\u21A3\u21A4\u21A5\u21A6\u21A7\u21A8\u21A9\u21AA\u21AB\u21AC\u21AD\u21AE\u21AF\u21B0\u21B1\u21B2\u21B3\u21B4\u21B5\u21B6\u21B7\u21B8\u21B9\u21BA\u21BB\u21BC\u21BD\u21BE\u21BF\u21C0\u21C1\u21C2\u21C3\u21C4\u21C5\u21C6\u21C7\u21C8\u21C9\u21CA\u21CB\u21CC\u21CD\u21CE\u21CF\u21D0\u21D1\u21D2\u21D3\u21D4\u21D5\u21D6\u21D7\u21D8\u21D9\u21DA\u21DB\u21DC\u21DD\u21DE\u21DF\u21E0\u21E1\u21E2\u21E3\u21E4\u21E5\u21E6\u21E7\u21E8\u21E9\u21EA\u21EB\u21EC\u21ED\u21EE\u21EF\u21F0\u21F1\u21F2\u21F3\u21F4\u21F5\u21F6\u21F7\u21F8\u21F9\u21FA\u21FB\u21FC\u21FD\u21FE\u21FF\u2200\u2201\u2202\u2203\u2204\u2205\u2206\u2207\u2208\u2209\u220A\u220B\u220C\u220D\u220E\u220F\u2210\u2211\u2212\u2213\u2214\u2215\u2216\u2217\u2218\u2219\u221A\u221B\u221C\u221D\u221E\u221F\u2220\u2221\u2222\u2223\u2224\u2225\u2226\u2227\u2228\u2229\u222A\u222B\u222C\u222D\u222E\u222F\u2230\u2231\u2232\u2233\u2234\u2235\u2236\u2237\u2238\u2239\u223A\u223B\u223C\u223D\u223E\u223F\u2240\u2241\u2242\u2243\u2244\u2245\u2246\u2247\u2248\u2249\u224A\u224B\u224C\u224D\u224E\u224F\u2250\u2251\u2252\u2253\u2254\u2255\u2256\u2257\u2258\u2259\u225A\u225B\u225C\u225D\u225E\u225F\u2260\u2261\u2262\u2263\u2264\u2265\u2266\u2267\u2268\u2269\u226A\u226B\u226C\u226D\u226E\u226F\u2270\u2271\u2272\u2273\u2274\u2275\u2276\u2277\u2278\u2279\u227A\u227B\u227C\u227D\u227E\u227F\u2280\u2281\u2282\u2283\u2284\u2285\u2286\u2287\u2288\u2289\u228A\u228B\u228C\u228D\u228E\u228F\u2290\u2291\u2292\u2293\u2294\u2295\u2296\u2297\u2298\u2299\u229A\u229B\u229C\u229D\u229E\u229F\u22A0\u22A1\u22A2\u22A3\u22A4\u22A5\u22A6\u22A7\u22A8\u22A9\u22AA\u22AB\u22AC\u22AD\u22AE\u22AF\u22B0\u22B1\u22B2\u22B3\u22B4\u22B5\u22B6\u22B7\u22B8\u22B9\u22BA\u22BB\u22BC\u22BD\u22BE\u22BF\u22C0\u22C1\u22C2\u22C3\u22C4\u22C5\u22C6\u22C7\u22C8\u22C9\u22CA\u22CB\u22CC\u22CD\u22CE\u22CF\u22D0\u22D1\u22D2\u22D3\u22D4\u22D5\u22D6\u22D7\u22D8\u22D9\u22DA\u22DB\u22DC\u22DD\u22DE\u22DF\u22E0\u22E1\u22E2\u22E3\u22E4\u22E5\u22E6\u22E7\u22E8\u22E9\u22EA\u22EB\u22EC\u22ED\u22EE\u22EF\u22F0\u22F1\u22F2\u22F3\u22F4\u22F5\u22F6\u22F7\u22F8\u22F9\u22FA\u22FB\u22FC\u22FD\u22FE\u22FF\u2200\u2201\u2202\u2203\u2204\u2205\u2206\u2207\u2208\u2209\u220A\u220B\u220C\u220D\u220E\u220F\u2210\u2211\u2212\u2213\u2214\u2215\u2216\u2217\u2218\u2219\u221A\u221B\u221C\u221D\u221E\u221F\u2220\u2221\u2222\u2223\u2224\u2225\u2226\u2227\u2228\u2229\u222A\u222B\u222C\u222D\u222E\u222F\u2230\u2231\u2232\u2233\u2234\u2235\u2236\u2237\u2238\u2239\u223A\u223B\u223C\u223D\u223E\u223F\u2240\u2241\u2242\u2243\u2244\u2245\u2246\u2247\u2248\u2249\u224A\u224B\u224C\u224D\u224E\u224F\u2250\u2251\u2252\u2253\u2254\u2255\u2256\u2257\u2258\u2259\u225A\u225B\u225C\u225D\u225E\u225F\u2260\u2261\u2262\u2263\u2264\u2265\u2266\u2267\u2268\u2269\u226A\u226B\u226C\u226D\u226E\u226F\u2270\u2271\u2272\u2273\u2274\u2275\u2276\u2277\u2278\u2279\u227A\u227B\u227C\u227D\u227E\u227F\u2280\u2281\u2282\u2283\u2284\u2285\u2286\u2287\u2288\u2289\u228A\u228B\u228C\u228D\u228E\u228F\u2290\u2291\u2292\u2293\u2294\u2295\u2296\u2297\u2298\u2299\u229A\u229B\u229C\u229D\u229E\u229F\u22A0\u22A1\u22A2\u22A3\u22A4\u22A5\u22A6\u22A7\u22A8\u22A9\u22AA\u22AB\u22AC\u22AD\u22AE\u22AF\u22B0\u22B1\u22B2\u22B3\u22B4\u22B5\u22B6\u22B7\u22B8\u22B9\u22BA\u22BB\u22BC\u22BD\u22BE\u22BF\u22C0\u22C1\u22C2\u22C3\u22C4\u22C5\u22C6\u22C7\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